# SEPNET PAID SUMMER PLACEMENT OPPORTUNITIES 2026 – UNDERGRADUATE

SEP net

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# Introduction and Contacts

Dear SEPnet Student,

We are delighted to be able to present the opportunities available for SEPnet work placements for Physics and Maths students. Please read through the list of projects carefully – they offer a great opportunity for you to gain valuable work experience this summer.

These projects are only open to some students and contain confidential information.

Please do NOT circulate this document to anyone else.

Details about the scheme are set out in the FAQs section below. Please make a note of the deadline dates, in particular, the application deadline of Monday 31 March.

If you have any questions, contact your Employer Engagement Adviser (EEA).

Details of your EEA at your university:

| | | |
|---|---|---|
| University of Southampton | Nicky Mlynar | N.S.Mlynar@soton.ac.uk |
| Royal Holloway | Jana Checkley | Jana.Checkley@rhul.ac.uk |
| QMUL | Amy Carroll | amy.carroll@qmul.ac.uk |
| University of Hertfordshire | Janice Hart | j.hart@herts.ac.uk |
| University of Portsmouth | Melvyn Vopson | melvin.vopson@port.ac.uk |
| University of Sussex | TBC* | careerhub.sussex.ac.uk |

*Jessica Eustace from the Sussex Central Careers Team can provide application support - book with her on email above

We wish you all the best with your applications!

# FAQ's: What You Need to Know

## How much and when will I get paid?

Successful candidates will receive a minimum payment of £3,560 gross which is for eight weeks work in the summer holidays. PhD students will receive a minimum payment of £3700 gross. If you are employed by the organisation you may be paid at the end of the month. If you receive payment via bursary via your home university you will be made aware of when this bursary payment will be made (likely to be in two stages). If you need payment early to cover expenses, let your host organisation know or contact your EEA for advice.

## How does the scheme work?

You need to register your details before you start applying. Click on the link here to register:

Undergraduates to register using this link:
https://www.smartsurvey.co.uk/s/UGRegisitration2026/

PhD students to register using this link:
https://www.smartsurvey.co.uk/s/PGRRegistration2026/

**NB: You will not be able to take up a placement if you have not registered here.**

1. Read the project descriptions in this booklet carefully. We recommend you apply for more than one placement and you may wish to apply for several but remember to target your applications to the projects that really interest you. Some placements may be located further afield. You should not let this put you off applying as you may lose out on a valuable opportunity. However you should make sure you can get there or be prepared to find accommodation nearby.
2. Visit your Careers Service for advice on your CV, covering email and preparing for interview.
3. Send a copy of your generic CV to your EEA for their records.
4. Apply direct to the contact given for each placement that interests you by the 31st March at the latest with a brief tailored covering letter and CV. Send these as a PDF with the company and your name eg Company_Surname_CV.pdf and Company_Surname_CL.pdf. Remember to be clear which placement you are applying for (in case there is more than one). It is better to apply early so your application stands out and to show you are keen. **Please note that many employers will start shortlisting and may recruit before the deadline!**
5. The covering letter should explain why you have chosen that placement, why you feel you are suitable for the role and when you are available for interview and work. Make sure your CV fits with what the employer is looking for. **Remember to include details of courses you have taken (avoiding abbreviations and highlight any good grades).**

6. Each organisation has its own recruitment process. Some may invite you to interview, some may ask you for a telephone interview, some may select on the basis of CV alone.
7. Organisations should contact you by end of April to let you know if you have been successful. It is perfectly acceptable to contact an employer if you have not heard back from them. If you are offered and accept a placement, you should honour this commitment and not change your mind at a later stage without good reason. If you have more than one offer and are unsure of what to do, contact your EEA for advice. **NB: YOU SHOULD ONLY ACCEPT ONE PLACEMENT AND CARRY OUT ONE SEPNET SUMMER PLACEMENT PER YEAR.**
8. Please see above regarding payment of a bursary. If you do not receive your payment, contact your EEA for advice. If you do not receive salary payment by your host organization you should speak 1st with your placement host/supervisor.

## Who is eligible?

You are eligible for the scheme if you are a physics or maths undergraduate in your 2nd or 3rd year and Integrated masters students (not MSc). 1st year students may also apply.at some universities (check with your EEA). SEPnet physics PhD students are also eligible to apply for some placements.

## I am an international student – am I able to take part?

Yes! While certain organisations can only offer places to British Nationals, the majority of placements are open to all. The project descriptions should state the cases where there are specific requirements. If you are uncertain of your status, check with your Student Services.

## What financial help is available?

The university or employer provides a payment of a minimum of £3,560 gross for eight weeks. This is intended to cover all your expenses over that time. Regrettably we cannot offer reimbursement for the following:

- Travel to interview (however it's worth asking your host company because some organisations may pay interview expenses).
- Accommodation expenses (your payment should help to cover these costs).

If you are paid monthly in arrears by your host organisation, you may be able to receive an advance payment to help cover accommodation costs. Ask your host supervisor or your EEA.

## What are the deadlines?

You need to complete and submit all of your applications by midnight **31st March**. You may then be called to interview or have a telephone interview. If you have been successful in securing a placement, you should find out by **end of April.**

## How do I find out further information?

If you have any enquiries contact your EEA or email summerplacements@sepnet.ac.uk

## Is there help with finding accommodation?

Some placements will require you to find temporary accommodation. Ask your host organisation for advice on finding accommodation and transport. You can also register with https://www.unihomes.co.uk/ or http://www.spareroom.co.uk/ . StuRents advertises properties in several regions throughout the UK (https://sturents.com/ ) and lists a wide range of summer lets. It also has a live support service during the day to help students find short-term accommodation over the summer period.

Contact summerplacements@sepnet.ac.uk if you want to be put in touch with other current or previous placement students.

## SEPnet summer placement timeline

- **1st March -** Projects are advertised by SEPnet Employability Advisers (EEAs) to students via email and other departmental communications.
- **Mid-March -** Employers provide feedback on the number and quality of applications.
- **31st March -** Closing date for applications.
- **1 – 30 April -** Employers select and interview candidates. Unfilled placements are re-advertised.
- **Early May -** Deadline for positions to be filled. Employers inform SEPnet and successful and unsuccessful candidates..
- **June – September -** Students carry out placements. Students prepare posters for Students' Expo. Placement visits or Teams call with students and supervisors take place mid-way through the placement.
- **November -** Students and supervisors participate in the SEPnet Students' Expo.

# Placements

## Aquark Technologies - Engineering | Hampshire | On-site

### Organisation Address

18 Hathaway Close, Eastleigh, Hampshire, SO50 4SR

### Start and End Date

June/July- Aug/Sept (flexible)

### Working Arrangements

In office preferred

### Company Description

Aquark Technologies, a start up from the University of Southampton, is working on miniaturising cold atom quantum technology for use outside of the lab. Cold matter devices use lasers to cool atoms to microkelvin temperatures in a vacuum but are often fragile, large, and power hungry. This limits the number of applications possible with these devices, therefore by miniaturising cold atom systems we can overcome these limitations. High performance and often bespoke hardware need to be developed to unlock the extremely precise measurement techniques possible with cold atoms to measure things like time, gravity, magnetic fields and acceleration. This means the company works across a large range of disciplines, including physics, electronics, photonics, software, mechanical and system engineering.

### Project Title 1: Intuitive Interfaces for Quantum Technology

### Your Contact

Name: Chester Camm
Email: c.camm@aquarktechnologies.com

### Project Brief / Job Description

Aquark Technologies has been developing the next generation of cold atom atomic clocks for commercial application. Part of this process has been to enable “non-experts” to successfully operate these devices – something usually only achieved by PhD level personnel. In order to fully enable anyone with brief training to operate these devices, Graphical User Interfaces (GUIs) are preferred. These can be designed so that the user is able to intuitively navigate the functionality of the device. This is a process that requires time,

patience, and an understanding of what a potential user might understand intuitively are required to successfully design and create good GUIs. We propose that these GUIs are designed and implemented.

## Student Specification

- Strong software development experience – Python preferred
- Able to work independently
- An interest in quantum technology

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## Project Title 2: Integrated Photonics for Quantum Technologies

## Your Contact

Name: Lauren Reid
Email: L.Reid@aqrkt.com

## Project Brief / Job Description

Aquark Technologies builds and delivers the world's smallest and most deployable cold-atom quantum hardware to enhance navigation, communications, security, and our understanding of the world around us.

Part of developing this technology involves the use of integrated photonics (or silicon photonics). This is an up-and-coming field that has experienced significant growth over the past couple of decades. It focuses on the creation of micron-scale optical components and complete systems on chips roughly the size of a fingertip. The fabrication methods of these chips are highly scalable, allowing many identical chips to be produced simultaneously. As a result, this technology is now key to applications such as data centres, LiDAR for self-driving cars, and AI computation.

In this project, you will experience the full design cycle of components on a photonic integrated chip. This includes simulation, mask design, fabrication, and an emphasis on testing and characterising components. The project will be varied and hands-on, requiring the use of Ansys Lumerical software to simulate components, gdsfactory (a popular Python package) for mask design, and laboratory testing of fabricated devices.

## Student Specification

- Confident in performing and analysing experiments
- Understanding of the physics of waveguides, fibres, or photonics
- Eager and enthusiastic to learn
- Desirable but not essential: coding experience (Python preferred)

# BEAT (Better Environment And Transport) - Transport, Research, Environment, Technology | Brighton | Remote

## Your Contact

Name: Matthew Dear
Telephone: 07721 870192
Email: Matthew.Dear@beatzero.co.uk

## Organisation Address

A Brighton based company BEAT | Better Environment And Transport

## Start and End Date

June- September (flexible). This is a 4 week placement with potential to extend subject to funding.

## Working Arrangements

Fully remote

## Company Description

BEAT develops AI and digital solutions to help organisations meet demanding Net Zero targets and support the decarbonisation of the Transport Sector.

BEAT works with clients who have demanding operational requirements - from fire and rescue services that must maintain emergency response times, to logistics companies with complex delivery schedules.

Our main solution is ZEEFLEET: an AI and optimisation platform that helps organisations provide clear reporting through integrating data across multiple systems, manage their operations more efficiently, and planning their transition to Zero Emission vehicles. ZEEFLEET combines operational data analytics, route optimisation, and predictive modelling to help organisations reduce costs today and identify the most effective pathway to electric fleet technologies and charging infrastructure.

As a Brighton-based technology company, we're focused on delivering practical AI solutions that create measurable environmental and operational benefits, helping the transport sector play its part in achieving Net Zero emissions targets.

## Project Title

### ZEEFLEET – Further Research and Development into AI and Optimisation Techniques for the Transport Sector

## Project Brief / Job Description

ZEEFLEET is an AI and optimisation platform helping emergency services and commercial fleets plan, manage and transition to Zero Emission vehicles. The platform combines operational data, route optimisation, and predictive analytics to help organisations reduce costs today while planning their shift to electric fleets.

This placement will enhance ZEEFLEET's analytical capabilities in three key areas:

1. Improving data visualisation to communicate complex fleet performance metrics to non-technical decision-makers
2. Exploring advanced forecasting models that account for seasonal variation geography, and climate impacts on both incident patterns and vehicle performance
3. Identifying and testing enhanced optimisation techniques for fleet deployment and charging infrastructure planning.

The student will work with real (anonymised) operational datasets from commercial and/or emergency fleets, gaining hands-on experience applying physics and mathematics to real-world sustainability challenges.

This is an exciting opportunity to enhance your analytical skills, gain industry experience, and work develop AI solutions that have real benefit and impact in the wider world.

## Student Specification

We are looking for a student that has all or some of the following skills and/or experience:

- Data visualisation
- Simulation
- Machine Learning
- Optimisation techniques
- Programming experience
- Interest in AI techniques for data analysis – ideally experience in the application of AI techniques
- Ability to communicate complex data into impactful, key outputs
- Ability to work independently and as part of a remote team

# CQER-IQ - Research/Quantum Physics/Energy Technologyy | London | Remote

## Your Contact

Name: Richard H Harris
Telephone: 020 3205 7018
Email: richardhharris@richardhharris.com

## Organisation Address

Level 1, Devonshire House, 1 Mayfair Place, Mayfair, London W1J 8AJ

## Start and End Date

Flexible (4 weeks) possible 8 weeks for RHUL candidates

## Working Arrangements

Project 1: Remote, or onsite at Royal Holloway (subject to confirmation)
Project 2: Fully Remote

## Company Description

CQER-IQ is an AI-augmented research lab conducting frontier research across multiple scientific domains. Founded by Richard H Harris MBE (BSc Theoretical Physics, Royal Holloway 1992), the lab operates at the intersection of theoretical physics, computational modelling, and AI-accelerated research. Active research streams include fusion energy (algorithm validated on UKAEA MAST data), next-generation batteries, cancer biology, quantum gravity experiments using Bose–Einstein condensate systems, and quantum photonics. CQER-IQ collaborates with world-leading research groups including Royal Holloway (Dr Richard Howl), UCL Physics (Professor Sougato Bose), the University of Portsmouth (Dr Sravan Kumar), the University of Southampton, Imperial College London, and CTU Prague. All research follows a structured methodology combining domain expertise with advanced AI systems, progressing from hypothesis through computational validation and laboratory partnership to commercialisation. Spin-out revenues fund the next wave of research.
See https://cqer-iq.com/ for full details.

## Project Titles

### Project 1: From Coupled Oscillators to Quantum Entanglement in Bose–Einstein Condensates

### Project 2: From Brownian Motion to Diffusion Processes

## Project Brief / Job Description

**Project 1: From Coupled Oscillators to Quantum Entanglement in Bose–Einstein Condensates**

This project builds from foundational quantum mechanics to a modern open question: can gravity mediate entanglement between massive quantum systems? The student will progress through coupled harmonic oscillators, normal modes, and quantisation to the physics of Bose–Einstein condensation. They will then computationally model a dual-BEC system, implementing and extending simulations of the Duan–Simon entanglement witness as a function of inter-trap separation, incorporating realistic noise models. This directly supports an active experimental programme designing laboratory tests of gravity-mediated entanglement (the Bose–Marletto–Vedral proposal). The emphasis is on building from first principles, asking curious foundational questions, and contributing to a genuine modern research problem. The work is entirely computational and literature-based, using Python and standard numerical libraries (NumPy, SciPy, matplotlib). No lab access is required. The student will produce a computational notebook and literature review suitable for poster presentation at the SEPnet Students' Expo. Supervised by Dr Richard Howl (Royal Holloway), with academic oversight from Dr K Sravan Kumar (Portsmouth) and Professor Sougato Bose (UCL)

**Project 2. From Brownian Motion to Diffusion Processes**

This project explores how macroscopic diffusion equations arise from the microscopic description of particles in Brownian motion, as explained by Einstein in 1905. The student will perform pedagogical research on how classical and quantum diffusion models are widely applied across STEM disciplines and have led to practical applications. As a particular example, they will numerically solve the basic setup of diffusion equations and ionic mobilities involved in the electrochemical processes that play a key role in lithium-ion and sodium-ion batteries. This connects fundamental statistical mechanics to a pressing engineering challenge in next-generation energy storage. The project involves multidisciplinary aspects of applied mathematics integrated with numerical study using Python or any other tool the student is familiar with. No lab access is required. The student will produce a computational notebook with working simulations and a literature review suitable for poster presentation at the SEPnet Students' Expo. Supervised by Dr K Sravan Kumar (University of Portsmouth, Royal Society Newton International Fellow 2022–24), who designed the project framework. This is genuine research connecting first-principles physics to commercially relevant battery technology.

## Student Specification

**Project 1**: 2nd or 3rd year physics student with competence in Python programming and a solid grounding in quantum mechanics. Familiarity with numerical methods (NumPy, SciPy) desirable. Interest in quantum foundations, gravity, or condensed matter physics beneficial.

**Project 2**: 2nd or 3rd year physics or maths student with competence in Python or equivalent computational tools. Grounding in classical mechanics and statistical physics desirable. Interest in energy, materials science, applied mathematics, or computational physics beneficial.

# Diamond Light Source - Research | Oxfordshire | On-site

## Your Contact

Name: Dr Genoveva Burca
Telephone: 01235 778281
Email:Genoveva.Burca@diamond.ac.uk

Proposed Supervisor: Dr Thomas Zillhardt

## Organisation Address

Diamond Light Source, Harwell Science and Innovation Campus, Didcot, OX11 0DE

## Start and End Date

Between 29th June -21st August (an 8 week project)

## Working Arrangements

In Office

## Company Description

Diamond Light Source is the UK's national synchrotron, a third-generation source that delivers bright, tuneable X-rays for cutting-edge research across the sciences. Co-funded by STFC and the Wellcome Trust, the facility operates 32 beamlines that serve as state-of-the-art laboratories for both academic and industrial users.

Beamline I12-JEEP, part of the Imaging and Microscopy Group within the Physical Sciences Division, provides high-energy X-ray imaging and diffraction for the non-destructive investigation of complex materials. Designed for in-situ and operando experiments, I12 supports studies of materials and components under realistic mechanical, thermal, chemical, or environmental conditions. This capability allows users to visualise structural evolution, damage mechanisms, phase transformations, and functional performance as they occur.

The beamline is widely used for research on industrial and engineering materials—such as titanium turbine blades, silicon carbide composites, steels, and advanced ceramics—alongside applications in plant sciences, archaeology, and cultural heritage.

Recent developments have expanded I12's capabilities, including improved propagation-based phase-contrast imaging for enhanced visualisation of low-density or weakly absorbing samples, and new methods for high-energy fluorescence tomography, enabling three-dimensional mapping of elemental composition alongside structural information. These advances support an increasingly diverse scientific community and broaden the beamline's impact across disciplines.

## Project Title

### AI-Assisted Data Fusion and Segmentation for Full-Field Fluorescence Tomography on the I12 Beamline

## Project Brief / Job Description

This 8-week project gives the student the rare opportunity to work with a newly installed, first-of-its-kind full-field fluorescence tomography setup on the I12-JEEP beamline at Diamond Light Source. Unlike conventional fluorescence tomography systems, which require slow raster scanning, the I12 configuration enables simultaneous acquisition of fluorescence and propagation-based phase-contrast tomography (PCT) using a large, unfocused beam. This unlocks genuinely 4D, element-specific imaging of materials and samples under realistic in-situ conditions.

The student will focus on developing and testing AI-assisted methods for merging fluorescence and PCT datasets, with an emphasis on improving spatial alignment, denoising, and segmentation of elemental distributions. Tasks may involve implementing or adapting machine-learning models for multimodal data fusion, training segmentation networks, and evaluating reconstruction pipelines using both simulated and experimentally acquired datasets. A short beamtime session will be included, during which the student will participate in operating the beamline to collect test data.

The placement is ideal for students with a keen interest in imaging, AI, or computational physics, and will provide hands-on experience with Python-based data analysis, synchrotron imaging techniques, and high-performance scientific workflows. By the end of the project, the student will prepare and present a report and prototype analysis tools demonstrating their approach, as well as a poster for presentation at the SEPnet conference.

## Student Specification

The project is well-suited for students interested in materials science, diffraction, AI, and energy research, with strong potential for publishable results.

The student should have a basic understanding of synchrotron radiation and Python programming. Further training may be provided in advance of and/or during the placement, where required.

In advance of the interview, it is encouraged that the student takes a look at the Diamond Light Source and I12 webpages, including recent publications, and it is also recommended that the student familiarise themselves with widely used AI tools for image processing and data analysis, including PyTorch and U-Net–based architectures.

# The Ditchley Foundation - Non-profit / NGO | Oxfordshire | Remote

## Your Contact

Name: Martin Smith
Email: martin@ditchley.com

## Organisation Address

Ditchley Park, Enstone, Chipping Norton, Oxfordshire, OX7 4ER, United Kingdom

## Start and End Date

The successful application can choose to start on either the 15th or 29th of June.

This will be a 6 weeks with a possible extension to 8 weeks.

This could be a part-time placement with the successful applicant working with us 3 days a week. To be discussed at interview.

## Working Arrangements

The internship will be fully remote. Note that there will be the option to visit Ditchley Park for a celebration event at the end of the internship (set for the 31st July to 1st August).

## Company Description

The renewal of democratic societies, states, markets and alliances is an urgent priority and action is needed from us all. The Ditchley Foundation aims to build new common ground so that we can move toward solutions to complex problems. We believe that change is driven, and managed, by personal relationships, not reports or declarations. We engage local and global communities over time rather than chase headlines. We are not tied to the agenda of any government, political party or commercial interest. In addition to our gold-standard conferences, which have taken place since the 1960s and have engaged decision-makers across government, finance, technology, academia and civil society, we regularly engage individuals across political stripes, expertise, generations and geographies in a series of small to medium-scale discussions. For Ditchley to be successful, it needs a deep understanding of the networks around a given theme, both from within its own network and the world outside. We are developing a suite of sophisticated tools to provide this understanding and to manage our network

## Project Title

### Bridging Divides with Graph Networks

## Project Brief / Job Description

This project leverages knowledge graphs to map and understand the human networks driving change in geopolitics, climate, technology and democracy. You will join a research team using qualitative and quantitative analysis to visualise how key individuals connect and influence these domains. You will critically explore the network's landscape, identifying structural strengths and hidden gaps in representation. Your insights will ensure Ditchley's 100+ annual events achieve the precise mix of diverse voices needed to spark innovation and tackle global challenges.

Further details can be found in the accompanying internship brochure https://tinyurl.com/ditchley-sepnet-ad-2026 and applications should be submitted to https://tinyurl.com/ditchley-sepnet-2026 no later than 31st March.

## Student Specification

We are looking for applicants with a curious mind who are eager to immerse themselves in the ethos of the Foundation. Our knowledge graph is hosted by Neo4j, which is easy to interface with (very similar to SQL) and so no previous experience with Neo4j or Cypher is required. Applicants must have competent quantitative skills, but these could be demonstrated in various ways, for example in coding ability (e.g. Python or R) or experience with statical software (e.g. SPSS). Note that we do not expect all of the above - these just give a flavour of the type of skills that we are looking for.

Although this project is better suited to undergraduate students, we are keen to hear from all students including PhD students who have an interest in the goals of the foundation and a desire to learn how data science skills can be applied outside of academia.

# Drumgrange - Engineering | Dorset and Surrey | Hybrid

## Your Contact

Name: Jestina Anderson
Telephone: 07825 462062
Email: janderson@drumgrange.com

**Supervisor Name**

Greg Mould/Andrew Bond

## Organisation Address

Chertsey Site: Drumgrange, Unit A, The Forum, Hanworth Ln, Chertsey KT16 9JX

Dorset site: Drumgrange, 15 South Way, Portland DT5 2NJ

## Start and End Date

June- Sept Flexible

## Working Arrangements

60% - Office

40% - Home

## Company Description

As a small-medium enterprise, we benefit from having a flat organisation structure with only a few levels between a junior role and a department head, so you can spend more time with technical application and less time wrangling with red tape. It also means that your voice is heard and there is plenty of opportunity for you to make a meaningful impact upon the business.

Roles here present fantastic opportunities to be involved across the whole design lifecycle, and with such a wide portfolio of products and capabilities, you can expect varied and career-expanding work that takes you across different engineering domains and environments.

The internship will be within the Science and Technology Team. This team has a number of experienced Physicists and Mathematicians with expertise in the development and implementation of sonar signal processing.

We welcome candidates from all backgrounds and particularly from sections of the community who are currently underrepresented within our industry, including women, ethnic minorities, people with disabilities and LGBTQ+ individuals. Please be aware that many of our roles will be subject to both security and export control restrictions. These restrictions mean that factors including your nationality, any previous nationalities held, and your place of birth may limit the roles you can perform in our organisation.

## Project Title 1: Development of Acoustic Transmission Model

## Project Brief /Job Description

Drumgrange is seeking a driven and resourceful intern to undertake acoustic model development activities at our facility in Portland, Dorset or our head office in Chertsey, Surrey. The structure of the internship may vary over the 8-week period depending on your interests and technical skills. The internship will predominantly focus upon the development of acoustic propagation models and verification through experimentation.

Whilst at Drumgrange the candidate would be expected to take part in a number of tasks from around the company and on various projects but their main focus will be within the Science and Technology (S&T) Team. The team comprises mathematicians and physicists working principally on sonar technologies.

Recent projects have focused upon the transmission of sound and vibration through structures. Drumgrange wish to expand upon this work and develop tools to automate predictions of how sound and vibration travel through large, complex structures. This will include direct sound transmission as well as leakage paths. Experimentation will be required and the intern will need to design and conduct the experiments with support from the S&T Team. Results will then be fed back into the model. A final report and presentation will be required.

The ideal candidate will be studying a STEM subject and have begun to develop experience in acoustics and signal processing.

## Student Specification

Undergraduate or Masters student for a technical degree subject such as, Physics, Maths, Electronics, Engineering or Digital Signal Processing.

- Strong analytical approach with problem-solving skills and an eye for detail.
- Interest in Applied Physics and Mathematical Modelling.
- Self-motivated to effectively carry out the work required to complete allocated tasks.
- Innovative and able to think outside the box.
- A team player who understands a proactive, collaborative approach.
- Solid experience in software programming, especially Python.

Due to the nature of our business, Drumgrange would require any applicant to obtain a basic security level of clearance in order to attend an 8-week placement with the Company. To obtain this you will need to be a UK National or UK Citizen with a minimum residency of 3 years living in the UK. Unfortunately, we are unable to offer a placement for any applicant who doesn't meet this criteria.

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## Project Title 2: Transient Acoustic Analysis and Prediction Internship

## Project Brief / Job Description

Drumgrange is seeking a driven and resourceful intern to undertake acoustic experimentation and signal processing activities at our facility in Portland, Dorset or our head office in Chertsey, Surrey. The structure of the internship may vary over the 8-week period depending on your interests and technical skills. The internship will predominantly focus upon the development of acoustic testing, analysis and predictive techniques utilising machine learning methods.

This project requires the applicant to have a strong background in data analysis. Some experience of acoustics, artificial intelligence or machine learning would be beneficial.

This project will involve the collection of acoustic data, and the further development and testing of an algorithm for detecting and classifying transient signals within the data. The applicant will need to design and conduct experiments to collect data, analyse the data, and begin the development of the classification tool. A final report and presentation will be required. The analysis and predictive tools will be developed in Python. All work will be done with the guidance of Drumgrange's experienced Science and Technology department.

This project will be the continuation of a project conducted by the S&T team supported by a SEPNet intern during 2025. The Intern was awarded second place for their poster describing their work on the project at the SEPNet Students' Expo.

The ideal candidate will be studying a STEM subject and have begun to develop experience in acoustics and signal processing.

## Student Specification

Undergraduate or Masters student for a technical degree subject such as, Physics, Maths, Electronics, Engineering or Digital Signal Processing.

- Strong analytical approach with problem-solving skills and an eye for detail.
- Interest in Applied Physics and Mathematical Modelling.
- Self-motivated to effectively carry out the work required to complete allocated tasks.
- Innovative and able to think outside the box.
- A team player who understands a proactive, collaborative approach.
- Solid experience in software programming, especially Python.

# Earth-i - Space Observation | London | Remote

## Your Contact

Name: Jennifer King
Telephone: 07480 024517
Email: jennifer.king@earthi.co.uk

## Organisation Address

Earth-i 3rd Floor, 86-90 Paul Street, London, EC2A 4NE, United Kingdom

## Start and End Date

Start between 1st & 15th June.

Between 20th July & 3rd August.

## Working Arrangements

The student will work remotely i.e. from their own home/campus accommodation, with daily communication with Earth-i colleagues on MS-Teams or Zoom. Earth-i is a fully remote company; however, it may be possible to organise in-person meetings to kick off the placement and later review the results. Details around travel arrangements and a suitable location for the meeting will be agreed with the student.

The student will need to use their own computer equipment. They will need their own good internet connection and suitable working space at home.

## Company Description

Earth-i's team of geospatial analytics professionals helps companies and governments make better decisions by providing them with geospatial insight, at global scale, simply and at low cost. We offer a range of services to suit each customer's needs; from consultancy to special projects, and from data sales to sophisticated information services that are easy for non-technologists to understand and use. We use data fusion to combine satellite operators, sensors and spectra to improve data frequency and accuracy.

We use Machine Learning & Computer vision to enable processing and insight generation on a global scale.

We transform images into insight, giving clients advanced data and charts rather than a multitude of images.

## Project Title

**Future Analytics for Nature through Observation and Modelling (FANTOM)**

## Project Brief / Job Description

Through the FANTOM project Earth-i is developing a high spatial and temporal resolution UK-centric database of agriculture and biodiversity markers and schemes.

The comprehensive database of markers and impact assessments will be made available to all governmental agencies, associated arms-length bodies and commercial companies, enabling them to monitor and measure the progress of their sustainability activities and interventions.

FANTOM aims to change the traditional approach of Earth observation-based land monitoring by not just looking at the asset (e.g. a land parcel) but also looking at its relationship with neighbouring assets, to support landscape-level sustainability activities across the wider landowning community.

You will join our application development team to carry out data processing and analysis in support of the FANTOM project. You'll work alongside experienced data scientists and remote sensing specialists to process satellite imagery, develop predictive models, and contribute to project deliverables including data products and associated documentation.

## Student Specification

This role would suit someone with an interest in satellite Earth Observation and the use of artificial intelligence techniques (computer vision, machine learning) to process it.

Some of the following skills would be an advantage:

- Basic experience with Python and libraries such as NumPy, Pandas, and scikit-learn
- Understanding of GIS tools (e.g., QGIS, ArcGIS)
- Familiarity with EO data formats (e.g., Sentinel, Landsat)
- Strong problem-solving skills and attention to detail

# Fat Fish Digital - Technology | Hybrid

## Your Contact

Name: Maddy Mossman
Telephone: 07387307014
Email: maddy@fatfishdigital.co.uk

## Organisation Address

5 Merchant Square W2 1AY

## Start and End Date

June-Aug  (Flexible)

## Working Arrangements

Hybrid (in-person days based in central London)

## Company Description

Fat Fish Digital is a technology consultancy and development studio with over ten years of experience at the forefront of digital solutions. We have delivered mobile apps and web-based platforms used by millions of people. We are a trusted technology partner for UK data-driven businesses.

## Project Title

### Development, Data, AI Summer Placement

## Project Brief / Job Description

We are looking for a motivated student to join us for a summer placement focused on development, data, and AI. This is a hands-on opportunity to gain real-world experience in software development, machine learning, and data analysis within a fast-moving tech environment. Responsibilities may include: Assisting in AI-driven product development, from research to implementation. Supporting web and mobile development projects using modern frameworks and tools. Working with data pipelines and analytics, helping to derive insights from real-world datasets. Collaborating with our developers, designers, data specialists, and AI engineers to solve technical challenges. Gaining exposure to agile development workflows and best practices in coding, product design, and AI.

## Student Specification

We are looking for a student with a passion for AI, data, and problem-solving. Strong analytical skills and a willingness to learn new technologies. We value initiative-takers with an ability to work independently and as part of a remote team. The ideal candidate would have some programming experience.

# Framestore - Film | On-site

## Your Contact

Name: Amy Smith
Email: amy.smith@framestore.com

**Please ensure you reference 'SEPnet' against your application. Include next to your name/and or in your covering letter. Applications must be submitted via the links below and NOT via a covering letter and CV to Amy.**

## Organisation Address

Framestore, 28 Chancery Lane, London, WC2A 1LB

## Start and End Date

06/07/2026 to 29/08/2026

## Working Arrangements

In office, 5 days per week (Mon-Fri), 9am to 6pm with an hour for lunch. Free accommodation is provided in central London for those who require it.

## Company Description

At Framestore, we bring extraordinary ideas to life in any space or screen through craft, imagination and technology.

Framestore is home to a collective of visionaries working across film, advertising, episodic and immersive. Our work and skillset is wide-ranging, thanks to a pool of 3000 talented artists and producers in offices across the globe.

## Project Title

### Summer Internship

## Project Brief / Job Description

Framestore's 2026 Launchpad Internship offers talented and creative individuals a chance to gain hands-on experience at one of Europe's leading creative studios. Interns will learn through studio work, presentations, a personal project, and informal meetups while collaborating on real projects with experienced artists and technologists. Previous interns have contributed to shows like One Piece, Project Hail Mary & How to Train Your Dragon.

## Student Specification

- Ability to respond positively to feedback and iterate accordingly
- Good time management and ability to meet deadlines
- Clear communication skills and a collaborative mindset
- Willingness to learn new tools, workflows, and techniques
- You must be entering the final year of your qualification or have completed in the past 12 months
- You must be a UK citizen or have a current resident, student or graduate VISA for the duration of the internship
- You must be available from 6th July until 29th August 2026 inclusive

**Must apply via the relevant link:**

**Animation** - https://framestore.recruitee.com/o/animation-launchpad-internship-2026

**Compositing** - https://framestore.recruitee.com/o/compositing-launchpad-internship-2026

**Environments** - https://framestore.recruitee.com/o/environments-launchpad-internship-2026

**FX** - https://framestore.recruitee.com/o/fx-launchpad-internship-2026

**Houdini Generalist** - https://framestore.recruitee.com/o/houdini-generalist-launchpad-internship-2026

**Immersive Realtime** - https://framestore.recruitee.com/o/rt-gen-l26

**Immersive Technologist** - https://framestore.recruitee.com/o/immersive-tech-l26

**Junior Cloud Engineer** - https://framestore.recruitee.com/o/cloud-engineer-l26

**Junior ML Ops Engineer** - https://framestore.recruitee.com/o/ml-ops-l26

**Junior Realtime Artist** - https://framestore.recruitee.com/o/rt-artist-l26

**Visual Development** - https://framestore.recruitee.com/o/vis-dev-l26

# Indra Space - Space Engineering | Oxfordshire | Hybrid

## Your Contact

Name: Roman Haskett
Telephone: 01235 635190
Email: rchaskett@indracompany.com

## Organisation Address

Deimos Space UK Ltd, Airspeed 1, First Floor 151 Eighth Street, Harwell Oxford, Didcot, Oxfordshire OX11 0RL

## Start and End Date

Flexible (Start and finish between June and Sept 8 weeks).

## Working Arrangements

Hybrid – Minimum 2 days in office per week, preferably Tuesday & Wednesday.

## Company Description

Indra Space (formerly Deimos) is an organisation with 11 branches across five countries: Spain, Portugal, the United Kingdom, Italy, and Romania, employing approximately 1,000 people. The company delivers end-to-end space solutions, offering capabilities across several key sectors, including telecommunications, Earth Observation and Intelligence, Surveillance and Reconnaissance (ISR), satellite navigation, space safety, and space exploration.

This placement will be hosted within the Space Situational Awareness (SSA) department, based at the Harwell Campus in the United Kingdom. Within the SSA domain, Indra Space has capabilities spanning the full Space Surveillance and Tracking (SST) value chain. This includes the operation of its own global optical sensor network, Deimos Sky Survey, with installations located in Spain, Chile, and Australia.

Beyond operating this sensor network, Indra Space plays a leading role in, or actively contributes to, a wide range of national and international projects. These are often delivered in collaboration with organisations such as the European Space Agency, the UK Space Agency, the Defence Science and Technology Laboratory (Dstl), and other partners. The focus of these projects is to develop advanced technologies that strengthen both European and UK capabilities in the Space Safety domain, supporting the long-term sustainability of space activities and contributing to a safer and more sustainable space environment.

## Project Title

### Investigation of Satellite Manoeuvre Detection and Pattern of Life Methods for Space Situational Awareness

## Project Brief / Job Description

Maintaining accurate knowledge of where satellites are in orbit is essential for Space Situational Awareness. This information is used to support collision avoidance, safe spacecraft operations, and the long-term use of Earth orbit. A major challenge arises when satellites perform orbital manoeuvres that are not publicly announced or are difficult to identify from observations. If these manoeuvres are not detected, predictions of a satellite's future position can quickly become unreliable.

The aim of this project is to explore existing methods used to detect satellite manoeuvres and to identify a satellite's Pattern of Life (PoL). In this context, PoL refers to recognising typical satellite behaviour, such as how often manoeuvres occur, the types of manoeuvres performed, and how a satellite's orbit changes over time.

The project will involve a review of state-of-the-art methods and techniques manoeuvre detection and PoL analysis, with a focus on understanding how manoeuvres can be characterised and categorised. The student will compare a range of approaches, highlighting their strengths, limitations, and typical applications.

By the end of the project, the student will produce a clear summary of current methods for manoeuvre detection and PoL analysis, along with recommendations on promising techniques that could support future work in satellite tracking and space safety.

## Student Specification

Whilst the following knowledge/skills are preferable, students are not expected to be experts providing they have a good interest in the subject matter of:

- Orbital Mechanics
- Good research skills and critical thinking
- State Estimation Filters
- Time Series Machine Learning

# Insensys Ltd - Fibre Optics | Hampshire | On-site

## Your Contact

Name: Sylwia Srodek
Telephone: 023 8045 0550
Email: ssrodek@insensys.com

## Organisation Address

Insensys Ltd,
12 Manor Court, Fareham, Hants, PO15 5TJ

## Start and End Date

Flexible (8 weeks within June to Sept)

## Working Arrangements

In Office

## Company Description

Insensys design and manufacture Fiber Optic Sensing systems. Our systems are installed in large wind turbines to measure aerodynamic loads on blade, to optimise performance and power output.
Founded 20 years ago, with around 65 employees based near Fareham.
Placement will be within our multi-disciplinary Engineering team covering, Optoelectronics, Electronics, Mechanical, Firmware and Software design.
Placement could lead to further opportunities with Insensys.

## Project Title

### Condition Monitoring, Feasibility Studies

## Project Brief / Job Description

Insensys are developing our Sensor Technology for Condition Monitoring, based on our Fiber Optic technology which involves developing algorithms for processing sensor data to provide Condition Monitoring solutions for Wind Turbines.
Applications include: Blade Imbalance Detection; Blade Resonance Monitoring; and early onset structural defects detection. Insensys are developing products for Wind Turbines.
Through the summer we will be performing several qualification test programs of new measurement hardware, which students could support.

Activities will include:

- Supporting in-house test activities: demonstrating product capability and reliability
- Supporting external testing: shock and vibration through to destruction

Test activities will need to be documented, with a level of data analysis to validate results.

## Student Specification

This is a large project, with many activities suitable for Maths and Physics students.

# Institute of Cosmology and Gravitation - Astrophysics Research | Portsmouth | On-site

## Your Contact

Name: Steve Cunnington
Telephone: 023 9284 8484
Email: steve.cunnington@port.ac.uk

## Organisation Address

University of Portsmouth, Institute of Cosmology & Gravitation
Dennis Sciama Building, Portsmouth PO1 3FX
02392 845817

## Start and End Date

Flexible (8 weeks within June to Sept)

## Working Arrangements

In office but flexible/hybrid options can be discussed

## Company Description

The Institute of Cosmology and Gravitation (ICG) is an international centre of research excellence in cosmology, gravitation and astrophysics. Our Cosmology and Astrophysics research investigates the evolution of galaxies and their stars, the early universe and large-scale structure in the cosmos, as well as the energy content of the Universe and the nature of gravity. Through the work of the institute, we seek to understand the physics of the Universe and inspire the next generation of scientist through education, training and outreach. We also innovate using our advanced skills in mathematics and data science. 100% of ICG research was judged to be world-leading or internationally excellent in the REF2021 Research Excellence Framework. The institute hosts 22 academic staff, 18 research fellows and 26 PhD students. Scientists at the ICG play leading roles in major ongoing international collaborations and projects, e.g. DES, DESI, Euclid, Rubin, SKAO/MeerKLASS, LIGO and LISA.

## Project Title

### Revealing the Structure of the Universe Using Radio Telescopes (Cosmology Research Placement)

## Project Brief / Job Description

Large-sky surveys are mapping our Universe across billions of light-years, with the aim of testing the behaviour of gravity across distances yet to be explored. This will shed light on

phenomena such as inflation in the early Universe and also test if late-time accelerated expansion is caused by dark energy or instead, some modification to general relativity. MeerKAT, a state-of-the-art radio telescope and precursor to the Square Kilometre Array Observatory (SKAO), is conducting an unprecedented spectroscopic survey of the Universe. By observing radiation from neutral hydrogen at 21cm radio wavelengths, the underlying large-scale cosmic structure can be mapped in a process known as 21cm intensity mapping. In this project, the student will work with some of the latest data from MeerKAT and trial new techniques to reveal large-scale cosmic structure with improved precision. The project will provide experience working within an active cosmology research department (Portsmouth's ICG), and an international collaboration: the MeerKAT Large Area Synoptic Survey (MeerKLASS). The student will ideally have some experience/tuition in coding (preferably in Python) and an interest in astrophysics (modules in these subjects not required, but will be a bonus).

## Student Specification

Physics/Maths or related degree students. Experience/familiarity with coding (preferably in Python). Modules in astrophysics preferred, though not required.

# Institute of Physics - Charity | London | Remote

## Your Contact

Name: Paul Williams
Email: studentcommunity@iop.org

## Organisation Address

Institute of Physics, 37 Caledonian Road, Kings Cross, London N1 9BU

## Start and End Date

30 June – 4 August (start date flexible)

4th September (end date flexible)

Option for 4, 6 or 8 week placement upon discussion .We can consider and agree on a case by case basis upon completion of interview stage and offer provided.

## Working Arrangements

Remote working with some office attendance

## Company Description

The Institute of Physics (IOP) is the professional body and learned society for physics in the UK and Ireland, with an active role in promoting co-operation in physics around the world. We strive to make physics and physics rich careers accessible to people from all backgrounds. We are a charity and membership organisation with over 20,000 physicists in our community, including thousands of students.

Student Engagement sits within the Membership department at the IOP, providing resources and opportunities to support students and members to develop their knowledge, gain valuable skills and connect with the community. We organise activities and volunteering opportunities through our Student Community network for university students and graduates to engage with physics, the physics community and IOP projects. We provide opportunities to develop physics focused knowledge and skills for life, including teamwork, critical thinking, leadership, stakeholder engagement and communication.

## Project Title

### Student Engagement Coordinator

## Project Brief / Job Description

The Student Engagement Coordinator will support preparations for the IOP Student Engagement programme of events, grants, ambassador and society support schemes ahead of the 2026-7 academic year.

This role represents an excellent opportunity to build professional skills and experiences that will serve you in any future job, with a nationally recognised physics centred organisation delivering an inspiring mission - and a footprint covering the UK & Ireland. You will work with and support students and physicists from undergrads through post-graduate to early career physicists and help promote the values and work of the IOP.

Responsibilities include but are not limited to:

- Development of the IOP Campus Ambassador (CA) programme
- Drive recruitment of IOP Campus Ambassadors for new academic year
- Support coordination of Autumn induction talks and workshops to universities
- Support and coordinate other student volunteer activity teams across a range of projects
- Lead a review of our affiliated society scheme and associated support
- Improve reach of our ambassador and community networks
- Support training of new ambassadors
- Support the overall development of the physics student community network and generate ideas, resources and activities that can meet their needs
- Support relevant IOP departments e.g. public engagement on appropriate projects
- Develop website content, update budgeting documentation and deliver comms campaigns
- Review and analyse relevant data to evaluate and improve our work

## Student Specification

We are seeking candidates with good communication skills who are passionate about physics, personal and professional development, public engagement, outreach, volunteering and building community and networks. Universities, PhD supervisors, employers and commercial organisations frequently cite experience of outreach and extracurricular engagement as an attractive quality in employees and we enable our networks to develop these skills and experiences.
We are seeking a candidate who is:

- Dedicated to making physics an inclusive welcoming space for everyone
- Enjoys communication and science communication, and in supporting and fostering community and connections in physics and STEM
- Possesses strong Outlook skills and project coordination skills (preferred) i.e. through any formal or informal event organisation; and have some experience of outreach
- Able to take the initiative to drive improvements in our work and be prepared to champion the IOP’s values independently at events or in writing;

- Able to show awareness of our programme of work with students and the wider student physics ecosystem; and ideally have some involvement in IOP student engagement activities already
- An IOP member (essential – if not already then you can signup once you've applied for the role).

# Institute of Physics, Czech Academy of Science (Prague, CZ) - Research | Prague | Remote

## Your Contact

Name: Dr Ippocratis Saltas
Email: saltas@fzu.cz

## Organisation Address

Institute of Physics of the Czech Academy of Sciences
Na Slovance 1999/2, 182 00 Prague, Czechia

## Start and End Date

June- September (flexible)

## Working Arrangements

Remote

## Company Description

Founded in 1954, the Institute of Physics of the Czech Academy of Sciences is one of the largest institutes in the Czech Republic pursuing fundamental and applied research across a wide range of disciplines at the forefront of current research. It hosts a vibrant international atmosphere and several prestigious, large-scale European grants.

## Project Title

### Deep Learning Tools for Gravitational Wave Science

## Project Brief / Job Description

The discovery of gravitational waves has brought a revolution in modern physics, opening up a new observational window into the Universe. In this project, the student will work on the development of machine learning tools for the detection and statistical analysis of gravitational waves from the mergers of black holes or neutron stars, using cutting-edge deep learning tools in Python. The results will be relevant for the science of current and future interferometer (space) missions such as the LIGO and LISA and will use simulated and real gravitational waveform data. The student will gain hands-on experience with methods at the forefront of current research of both astrophysics and artificial intelligence. The prerequisite for the internship is a strong foundation of Python and PyTorch.

## Student Specification

Minimum requirements:

- Currently pursuing or recently completed a degree in Engineering, Physics, Computer Science, Mathematics, or a related field.
- Good foundation of coding in Python and/or PyTorch.
- Strong analytical and problem-solving skills with attention to detail.
- Excellent communication skills and the ability to present findings clearly.
- Eligible to work in the UK.
- Passionate about AI and a strong desire to learn and develop new skills.

# In2science - Education | London | Remote

## Your Contact

Name: James Cook
Email: recruitment@in2scienceuk.org

## Organisation Address

In2scienceUK, 605 Albert House, 256-260 Old Street, London EC1V 9DD

## Start and End Date

End of June – End of Aug (flexible 8 weeks)

## Working Arrangements

Mostly remote with the possibility of occasional in-person events

## Company Description

In2scienceUK is an award winning charity and our mission is to support young people from low socioeconomic backgrounds to achieve their potential and progress to science, technology, engineering and maths (STEM) degrees, apprenticeships and careers to become the inventors and pioneers of the future. To date the charity has supported thousands of young people through our impactful programmes including In2STEM which provides work placements, skills workshops and career guidance. In 2025 In2STEM supported over 600 students across the UK, connecting them with world-class scientists, engineers, technologists and researchers to develop the skills and confidence for a successful career in STEM. In 2026 In2scienceUK will be continuing to develop our In2STEM, In2research and alumni to support more people from low socioeconomic and under-represented groups than ever before.

### Project Title 1: In2STEM Programme Intern

## Supervisor Name

Jordan Stephenson

## Placement Description

This role will support the In2STEM programme team in delivery of the In2STEM programme. Responsibilities include: working with volunteers to moderate workshops; visiting students on placement; collecting and writing case studies for the website and social media; creating newsletters and drafting communications for students and volunteers; supporting the team

on other projects undertaken over the summer. There may be opportunities to be involved other aspects of the charity, depending on interest.

## Student Specification

You will be someone with an interest in promoting social mobility and diversity in STEM, who enjoys working with young people and has an understanding of the barriers facing young people from low socioeconomic backgrounds when accessing higher education.

You will also have strong organisation, prioritisation, communication and interpersonal skills.

Essential: Excellent communication and interpersonal skills that adapt to a range of audiences; highly organised and efficient, ability to work effectively without direct supervision, able to prioritise tasks; responds to situations calmly, flexibly and appropriately, displaying resilience particularly when faced with challenge.

Desirable: Competent user of G-Suite applications (Gmail, Docs, Sheets); experience using social media in a professional context; experience of using charity specific IT systems (Brevo, Monday.com, Zoom), experience of working with young people, particularly in a classroom/workshop/youth work environment; experience of supporting the delivery of events and/or complex projects involving multiple tasks.

# Project Title 2: In2research Programme Intern

## Supervisor Name

Mitzi Tahsin

## Placement Description

During the internship, the current 2025/26 In2research cohort will be on their research placements, with the delivery team supporting them through this and preparing for the 2026 Celebration Event, where the achievements of the In2scienceUK programmes and alumni will be celebrated. At the same time, the team will also be recruiting volunteers and participants for the 2026/27 In2research programme. Therefore, this role will support the In2research programme delivery team, with responsibilities possibly including: drafting communications for participants and volunteers; reviewing outputs to ensure they are in line with specifications; attending and writing up case studies from site visits; drafting and proofing communications for participant and volunteer applicants, for email, website and social media; assisting with event planning; completing administrative tasks such as inbox management and data management; supporting other projects undertaken over the summer.

There may be opportunities to be involved in other aspects of the charity, depending on interest.

## Student Specification

You will be someone with an interest in promoting social mobility and diversity in STEM, who enjoys working with young people and has an understanding of the barriers facing young people from low socioeconomic backgrounds when accessing higher education.You will also have strong organisation, prioritisation, communication and interpersonal skills.

Essential: Excellent communication and interpersonal skills that adapt to a range of audiences; highly organised and efficient, ability to work effectively without direct supervision, able to prioritise tasks; responds to situations calmly, flexibly and appropriately, displaying resilience particularly when faced with challenge.

Desirable: Competent user of G-Suite applications (Gmail, Docs, Sheets); experience using social media in a professional context; experience of using charity specific IT systems (Brevo, Monday.com, Zoom), experience of working with young people, particularly in a classroom/workshop/youth work environment; experience of supporting the delivery of events and/or complex projects involving multiple tasks.

# IsardSAT - Earth Observation | Surrey | Hybrid

## Your Contact

Name: Mark Pattle
Telephone: 01483 685580
Email: mark.pattle@isardsat.co.uk
Supervisor Name: Charlie McKeown

## Organisation Address

Surrey Research Park, Guildford, Surrey GU2 7YG, UK

## Start and End Date

June –Sept Flexible

## Working Arrangements

Hybrid (3 days per week in office)

## Company Description

isardSAT Group (also known as isardSAT) is a group of R&D SMEs providing, since 2006, engineering and scientific remote sensing answers to its clients with special attention to altimetry and passive microwave missions (such as altimetry, radiometry, and SAR images). isardSAT Group is composed of isardSAT, S.L., Barcelona headquarters and a fully owned subsidiary, isardSAT Ltd., in Guildford.

isardSAT is involved in the design, calibration, and maintenance of civil remote sensing instruments, ranging from the development of algorithms for data processing to calibration after launch and maintenance of the instrument's hardware until decommissioning, and is continuously improving its processing software well after that decommissioning phase.

isardSAT’s also performs studies for scientific applications with the data obtained by Earth Observation instruments, such as for hydrological, coastal, and cryosphere applications.

isardSAT has a wide experience successfully managing complex technical projects lasting several years as can be inferred from the increasing number of ESA-awarded projects since the company creation.

## Project Title

### Earth Observation: Satellite Data Processing and Analysis

## Project Brief / Job Description

The Space & Technology team at isardSAT designs and develops algorithms related to oceanography, hydrology, coastal, and cryosphere applications using microwave instruments (altimeters, radar imagers, and radiometers).

We are looking for a placement to support projects in the Earth Observation field with different levels of complexity and physical concepts to develop and validate new applications in the coastal, cryosphere, and inland waters domain.

The chosen task would depend on the candidate's interests and aptitudes in the potential options, as well as the company's priorities at the time of the internship.

Potential tasks include:

- Developing machine learning models to enhance our Earth Observation processing
- Researching and developing a mathematical model related to satellite Radar altimetry.
- Testing and improving existing methods in satellite Radar altimetry, specifically related to the Cryosphere.

## Student Specification

Knowledge, skills, and abilities:

- Student (Bachelor, Master, or PhD) in Maths, Physics or similar (Telecommunications Engineering)
- Interest in Remote Sensing and in Earth Observation
- Good knowledge of Python and/or MatLab
- The knowledge of other languages (such as C or C++) is considered an asset
- The knowledge of signal or image processing is considered an asset
- Knowledge of machine learning methods is considered an asset

The following personal skills are required:

- An aptitude for problem-solving.
- Self-motivated.
- Able to work as an individual and as part of a team of engineers and scientists, to communicate effectively with engineers a range of people, to work to tight deadlines in a logical manner and with attention to detail.
- A desire to learn and develop new skills, with the willingness to ask questions and the ability to learn and research independently when needed.

# MeVitae - Research, Engineering, Technology | Oxfordshire | Remote

## Organisation Address

MeVitae (Oxiway Limited), F26 R27 Atlas Building, Harwell OX11 0QX

## Start and End Date

July- September Flexible

## Working Arrangements

Remote

## Company Description

MeVitae's vision is to create fairness in the workplace, empowering organizations to make smarter, faster, and fairer decisions. Integrating fluidly with existing HR systems, MeVitae consolidates critical workforce data - from recruitment to payroll - into a single platform, proactively identifying human capital risks and challenges. Leveraging AI-driven personalized interventions, MeVitae orchestrates precise, tailored strategies to pre-emptively address these challenges. These interventions (such as anonymised recruiting) are pivotal in reshaping recruitment and employee management practices. Proven results underscore MeVitae's efficacy: achieving up to 90% in time and cost savings, reducing employee turnover by 25%, and mitigating risks by 30%. Esteemed by top-tier government bodies and leading global brands such as Southern Company, Transport for London, Hogarth Worldwide, and Taylor Wessing, MeVitae stands as a trusted partner in transformative workforce management.

As a close-knit and relaxed team, we're eager to welcome a new member who is as driven and ambitious as we are. If you're passionate about accelerating your career and advancing the cause of workplace fairness, MeVitae is your platform to thrive. Join us in shaping a fairer, more efficient workplace worldwide.

### Project Title 1: Back End Developer Intern

## Your Contact

Name: Jacob O'Toole/ Ethan Cooper
Telephone: 07796 533839
Email: careers@mevitae.com

## Project Brief / Job Description

Are you passionate about using your software engineering skills to make a significant impact? At MeVitae, we're not just coding; we're engineering a brighter future for the future of work. As a Backend Developer, you'll play a role in shaping our technology stack as we rapidly develop our products.

In this role, you'll be presented with a choice of projects utilising different backend technologies and choose the one you prefer based on your skills and what you'd like to learn - these are likely to be in Go, Rust, Python, or C#.

Your responsibilities will include:

- Engage in architectural design sessions and technical discussions for developing robust backend systems. Draft and review implementation plans to ensure efficient execution of projects.
- Take an active role in code review sessions by providing thoughtful and constructive feedback on other's pull requests, which quickly actioning feedback received on your own.
- Execute daily programming tasks with a focus on writing clean, efficient, and maintainable code using Go, Rust, C#, and/or Python.
- Collaborate in the formulation and revision of security policies to align with industry best practices. Help ensure that all your code adheres to established security protocols and guidelines to protect sensitive data and systems.

## Student Specification

- Programming Proficiency: Demonstrated expertise in development with Python, Go, Rush or C# – or any other language, with a demonstrated ability to learn and adapt quickly.
- Problem Solving and Teamwork: Excellent problem-solving abilities with the capacity to work independently and collaboratively within a team.
- Communication Skills: Effective communication skills, capable of clearly articulating technical challenges and solutions to a varied audience.

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## Project Title 2: Customer Success & Implementations Intern

### Your Contact

Name: Ellie Young / Ethan Cooper
Telephone: 07796 533839
Email: careers@mevitae.com

### Project Brief / Job Description

We are looking for a passionate and enthusiastic intern to join our Customer Success team. You will utilise your technical ability to support the Customer Success team and Integration specialist with improvements to their current software that includes Totango, Hubspot, Jira and Github. As part of your role, you will be given the opportunity to meet and liaise with our current clients, including a selection of the large enterprise organisations, providing you with great corporate exposure.
As a scaling start-up, we are a tight team who work closely with each other, but we not believe in micromanagement and so from day one, you will be given autonomy to develop your communication and time management skillsets. We're looking for someone with a deep desire to learn and gain extensive experience in an exciting, but fast pace, start-up environment. In exchange, we offer unlimited opportunities for you to grow and develop, exploring new paths as you kickstart your career.

Responsibilities

- Collaborate with the Customer Success team to understand the technical requirements for their current software solutions.
- Collaborate with the Senior Integrations Specialist to implement the enhancements to the software solutions.

### Student Specification

- Problem Solving and Teamwork: Excellent problem-solving abilities with the capacity to work independently and collaboratively within a team.
- Communication Skills: Effective communication skills, capable of clearly articulating technical challenges and solutions to a varied audience.
- Technical Ability: Experience in Python, or similar programming language with ability to show debugging skills. Clear understanding of how to use Dashboards and other Configuration tools. Basic concept of ticketing systems.

Good to have:

- Interest and knowledge of human behaviour and cognitive biases and/or AI (NLP). Awareness of current trends in ML/AI.

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## Project Title 3: Frontend Developer Intern

### Your Contact

Name: Jacob O'Toole/ Ethan Cooper
Telephone: 07796 533839
Email: careers@mevitae.com

### Project Brief / Job Description

As a Front End Developer you will play a pivotal role in advancing our platforms and solutions through the development of our dashboards. In this key position, you will gain experience in Next.js, React, and TypeScript to develop responsive and robust user interfaces that enhance user engagement and functionality across all devices. Collaborate with our talented team of UX/UI designers and backend developers you will ensure seamless integration and superior performance of our solutions. This role offers the opportunity to challenge your skills in a dynamic environment, support innovative projects, and develop applications that make a difference. Join us to shape the future of fairness with your expertise and insight.

MeVitae is proud to be an Equal Employment Opportunity employer. We believe passionately that employing a diverse workforce is central to our success. We do not discriminate based upon race, religion, colour, origin, sexual orientation, gender identity, gender expression, age, status as a protected veteran, status as an individual with a disability, or other applicable legally protected characteristics.

### Student Specification

- Testing and Debugging: Proficient in using tools and techniques for testing and debugging.
- Willing to learn new coding languages, practices, and tools.
- Proficient in Technologies: High-Level understanding of Python essential. Experience of HTML5, CSS3, and JavaScript favourable.
- Experience with Responsive Design: Demonstrable skills in creating fluid and flexible layouts that work across different devices and screen sizes.
- Version Control/Git: Ideal experience with version control systems, particularly Git, to manage changes to the source code.
- UI/UX Design Sensitivity: Good sense of design aesthetics and an understanding of user experience fundamentals. Ability to implement UI/UX designs with accuracy.

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## Project Title 4: NLP Intern

### Your Contact

Name: Jacob O'Toole/ Ethan Cooper
Telephone: 07796 533839
Email: careers@mevitae.com

### Project Brief / Job Description

We are seeking a passionate and motivated Statistics/Machine Learning Intern to join our team during this exciting period of growth. You will play a crucial role in researching, developing, and implementing machine learning models, with a particular focus on Natural Language Processing (NLP). This position offers a unique opportunity to work on innovative projects alongside experts in the field, while honing your skills in a dynamic and collaborative start-up environment. Your primary responsibilities will include designing and testing models, working with NLP algorithms to analyse and interpret textual data, and applying statistical techniques to solve real-world problems. You will explore state-of-the-art trends in machine learning and AI, particularly in areas such as language models and cognitive systems. In addition, you will have the opportunity to contribute to the development of scalable, efficient pipelines for data preprocessing and model deployment. While previous relevant experience or a related academic background would be advantageous, we are most interested in finding someone with a strong desire to learn, tackle complex challenges, and contribute meaningfully to our mission.

### Student Specification

We are looking for applicants with a keen interest in statistics, machine learning, with a strong desire to learn and develop new skills.

Self-motivated with good organization skills to adapt to a fast-paced environment with changing priorities. A desire to work in an exciting and innovative start up environment.

Since we're remote, we need excellent communication – both written and verbal.

Basic knowledge of/experience of programming (ideally in Python, though this is not strict) is required, along with awareness of current trends in ML/AI and ideally some experience training and evaluating machine learning models.

# NATS - Aviation | Fareham | Remote

## Your Contact

Name: Richard Everson
Email: r.m.everson@exeter.ac.uk

## Organisation Address

NATS Corporate & Technical Centre, 4000 Parkway, Whiteley, Fareham, PO15 7FL

## Start and End Date

Flexible. Note: This is a 3 month placement

## Working Arrangements

Mainly homebased, with occasional visits to University of Exeter, NATS (Southampton) and the Alan Turing Institute (London).

## Company Description

NATS (formerly National Air Traffic Services), is the main air navigation service provider for the UK. It provides air traffic control services over the UK and North Eastern Atlantic, as well as to most major airports including Heathrow.

NATS are undertaking a £15M EPSRC investment for Project Bluebird, a 5 year partnership with The Alan Turing Institute, which is the UK's national institute of data science and AI, to deliver the world's first AI system to control a section of airspace in live trials, as well as delivering a "Digital Twin of UK airspace", a data driven, probabilistic simulator of UK.

NATS website: NATS - A global leader in air traffic management and airport performance

## Project Title

### An Agent to Coordinate Aircraft Between Sectors

## Project Brief / Job Description

Project Bluebird (https://www.turing.ac.uk/science-innovation/defence-and-national-security/project-bluebird) is developing AI agents to capable of autonomously controlling aircraft within a sector of airspace. As aircraft transition from one sector to another the location and particularly the height at which the aircraft will make the transition must be negotiated between the sectors. The goal of this project is to develop an agent that will negotiate with its counterpart in another sector to find suitable transition points and levels.

We will begin with negotiating suitable levels using simple offer and response interactions and using the agents in each sector as “subroutines” to quantify how easy it would be to control an aircraft entering at the proposed level.  We will use the extensive Bluebird DT digital twin of UK airspace in which to develop the agent.

Right now, the microscope can only produce an electric field along one axis, limiting the defects that are responsive to it (and therefore, that we can see). You will simulate, build, and test enhancements to this microscope to reveal defects that are, as of yet, undetectable.

This is an exciting opportunity that offers a motivated student the ability to work in a world-leading research group, with your contributions being used directly to improve the microscope’s sensing abilities, ultimately informing fabrication decisions for an entire industry. Throughout the project you will gain hands-on experience working in a laboratory on low temperature physics, scanning gate microscopy, and quantum computing.

## Student Specification

- Computer literate
- Interest in Data Analysis
- Good attention to detail
- Good programming skills and an interest in game theory would be useful.

# NPL - Research, Engineering, Technology | London | On-site

## Your Contact

Name: Toby Hawkins
Telephone: 020 8943 6232
Email: toby.hawkins@npl.co.uk

## Organisation Address

National Physical Laboratory, Hampton Road Teddington, TW11 0LW

## Start and End Date

July- September Flexible

## Working Arrangements

In Office

## Company Description

The Quantum Technology department at the National Physical Laboratory, the UK national metrology institute, develops measurement techniques and metrology standards to support the development of quantum technologies in the UK. This project would take place within the Quantum Computing Circuits group, in a team researching measurement technology for cryogenic devices and circuits.

## Project Title

### Designing, Building, and Testing Enhancements to a Scanning Quantum Probe Microscope for Live Readout of Superconducting Circuits

## Project Brief / Job Description

A key challenge in realising quantum computing is the mitigation of chip defects that are a source of decoherence, limiting their performance and scalability. The Quantum Computing Circuits group at NPL has developed a novel microscope, operating at millikelvin temperatures, that is uniquely capable of imaging these defects on live superconducting circuits. The microscope works by applying an electric field to the tip of an atomic force microscope and measuring perturbations in the circuit caused by this field at different strengths and locations.

Right now, the microscope can only produce an electric field along one axis, limiting the defects that are responsive to it (and therefore, that we can see). You will simulate, build, and test enhancements to this microscope to reveal defects that are, as of yet, undetectable.

This is an exciting opportunity that offers a motivated student the ability to work in a world-leading research group, with your contributions being used directly to improve the microscope's sensing abilities, ultimately informing fabrication decisions for an entire industry. Throughout the project you will gain hands-on experience working in a laboratory on low temperature physics, scanning gate microscopy, and quantum computing.

## Student Specification

The student should have a strong interest in experimental physics with a good understanding of quantum mechanics and electromagnetism.

Experience with soldering, EM modelling via COMSOL/HFSS, and knowledge of superconductivity are desirable, but not required.

# Openshaw & Co - Legal | Surrey | Hybrid

## Your Contact

Name: Tim Beere
Telephone: 01252 741741
Email: tim.beere@openshaw-and-co.com

Supervisor
Andrew Hollos
andrew.hollos@openshaw-and-co.com

## Organisation Address

Work will be on hybrid working model (some office and some remote)
Openshaw & Co, 8 Castle Street, Farnham, Surrey. GU9 7HR

## Start and End Date

29th June – 21st August

## Company Description

Openshaw & Co is a UK and European Patent Attorney firm based in the heart of Farnham, Surrey. Our clients range from large multinationals to SMEs, universities and start-ups, covering a wide range of interesting and advanced technologies, including aerospace, telecommunications, energy, defence, security, computing, and software.

## Placement Title

**Trainee Patent Attorney**

## Project Brief / Job Description

This placement will give students experience of the work that a Patent Attorney undertakes. The work will include:

- the analysis of scientific or technical documents, including previously granted patents, to assess whether an invention is new and inventive;
- the preparation of responses to reports from patent examiners, containing detailed and logical argumentation; and
- the writing of detailed descriptions of inventions in precise legal terms (patent drafts).

The student will be exposed to a wide range of interesting and advanced technologies. The student will gain an insight into the IP profession and what is required to train and qualify as a Patent Attorney in private practice.

## Student Specification

The student must possess, or be in line to achieve, a minimum 2:1 Honours Degree in Physics, Mathematics, or Electronic/Electrical Engineering, or similar. The student should have a natural, inquisitive desire to advise clients in the protection of intellectual property rights. The student should possess an aptitude for communicating complex information in a straightforward manner, be confident, meticulous, and have a keen eye for detail.

# Penta Group - Data Science | London | Hybrid

## Your Contact

Data Science Lead: James McVey
Email: james.mcvey@pentagroup.com
CTO: Trevor WatkinsHybrid.  At least two days a week in the London office, the days the whole of tech goes into the office are Monday & Wednesday.
Email: trevor.watkins@pentagroup.com

## Organisation Address

7th Floor, 55 King William Street, London, England, EC4R 9AD

## Start and End Date

Start: Early – mid June 2026

## Working Arrangements

Hybrid.  At least two days a week in the London office, the days the whole of tech goes into the office are Monday & Wednesday.

## Company Description

Penta is the world's first comprehensive stakeholder solutions firm. Penta provides data-driven solutions for stakeholder strategy and engagement. We combine analytics and research with deep expertise to deliver actionable insights, best practice and solutions, and drive measurable impact for our clients.

At Penta, we work with clients to solve complex global challenges across a variety of industries. Our clients span technology, financial services, energy, transport, agri-food, and more. Our team brings decades of experience in business, government, communications, research, technology, and media.

Penta comprises over 350 professionals in New York, London, Washington, DC, San Francisco, Dublin, Brussels, Hong Kong and Singapore working to bring data-driven, tailored solutions to the clients.

## Placement Title

### Data Science Intern

## Project Brief / Job Description

The placement is provided in the Data Science team, which is part of our Technology team.

You will:

- Participate in the development of ***Generative AI / ML applications for the in-house framework of the AI Analysis Tools***, using Python and applying the latest advancements in Data Science to real-world use cases.
- Leverage Penta's internal big data sources of textual data.
- Work on several API that are the backbone of our AI offering
- Apply ***natural language processing/understanding, machine learning, and Large Language Models*** to develop AI solutions.
- Be a part of a Tech team.

## Prospective projects to work on (all Python):

- Synthetic Stakeholders – use our data to create Personas who can interact to take surveys, give opinions on political events or feedback on pitches and adverts
- Penta Platform GPT - interpret users natural language requests which will involve them defining a dataset, asking for various types of analysis and then presenting the output of the analysis in the chat
- Penta Brain – use LLMs for contextual analysis at scale, so we can make more of our data than we can with traditional insights methods
- Penta Office - Use of AI to create PowerPoint slides & Excel based on Penta data & client requirements

## Student Specification

Apply if you have …

- Passion for AI development and a fascination with data science
- Coding skills in Python for Data Science
- Some knowledge of Natural Language techniques
- Familiarity with Machine Learning basics
- Some knowledge of SQL to manipulate datasets
- Ability to research and gather information independently
- Ability to analyse and organise data (data handling skills)
- A desire to improve your technical skills, using a modern tech stack such as AWS, containerisation, FastAPI, Python, Postgres & some vibe coding

If you are missing some of the technical skills above but are keen to learn, please still get in touch!

Don't apply if you …

- Hate technical work
- Don't like 'getting your hands dirty'
- Don't like to learn new things on the fly

# QMUL - Research/Higher Ed | London | On-site

**(QMUL Students not eligible to apply)**

**QMUL (Queen Mary University London)**

## Your Contact

Name: Nicola McConkey
Telephone: 020 7882 5555
Email: n.mcconkey@qmul.ac.uk

## Organisation Address

Queen Mary University of London, G O Jones Building, 327 Mile End Rd, Bethnal Green, London E1 4NS

## Start and End Date

6 weeks flexible ( start between 1st July and 20th July )

## Working Arrangements

On site but can be flexible to accommodate hybrid.
University accommodation is sometimes available over the summer if booked sufficiently far in advance.

## Company Description

This project is hosted by the particle physics research group at Queen Mary University of London (QMUL), within the Department of Physics.

## Project Title

### Neutrino Physics with the Short-Baseline Near Detector (SBND)

## Project Brief / Job Description

The Short-Baseline Near Detector (SBND) is a neutrino detector based at Fermilab (USA), which is currently collecting data in a neutrino beam, but already has the largest global dataset of neutrino interactions on argon!

In this project you will work with SBND data and simulations to analyse neutrino interactions, for example using simulation to make a feasibility study for a future SBND measurement or working with the QMUL research group to characterise the current performance of the SBND detector.

There is also an option for involvement in a laboratory-based activity, doing research and development for a prototype detector.

## Student specification

This project will suit a student with coding skills in python or C++, excellent problem solving skills, and an interest in particle physics.

Involvement in the prototype detector development would suit a student with excellent problem solving skills and an interest in technical, laboratory based work.

# Qrometric - Metrology | Portslade, East Sussex | On-site

## Your Contact

Name: Callum Morgan
Telephone: 020 7099 5807
Email: callum@qrometric.com

**Supervisor name: Jack Maynard**

## Organisation Address

Qrometric Ltd, Unit 10, Hove Enterprise Centre, Portslade, East Sussex BN41 1UY

## Start and End Date

01/07/2026- 30/09/2026 (approximate and/or flexible 4 weeks, 8 weeks dependent on funding TBC)

## Working Arrangements

In office/laboratory

## Company Description

Qrometric is a UK metrology company specialising in precision instrumentation, product and system innovation.  Located on the waterfront at Portslade in Brighton & Hove, East Sussex, our team is dedicated to developing and supplying world class metrology products that address measurement challenges traditional technologies cannot solve.

Qrometric supports customers including national measurement institutes, calibration laboratories, pharmaceutical manufacturers, energy generation and transmission companies, and leading research facilities. Our expertise spans the full spectrum of humidity and temperature calibration, measurement optimisation, and instrumentation system development.

## Project Title

### Investigating the Effects of Ice Morphology for Measurement of Frost Point Using Chilled Mirror Hygrometers

## Project Brief / Job Description

You will be playing a leading role in investigating the effects of various ice phases, their formations, and their effects on measurement of frost point. This will be part of increasing the

stability accuracy of Qrometric's chilled mirror hygrometer, in temperature ranges of -80 to -100°C frost point. These measurement ranges are usually only achieved at national standards labs or metrological institutes. The project will also include opportunity for more novel solutions, such as using acoustic vibrations to control the formation of ice.

Working closely with our team and in collaboration with the national metrological institutes of Japan, Italy and the UK, you will analyse measurement data and identify potential areas for improvement. The nature of practical, coding and experimental work will depend on skillset and what you are comfortable with, but may include soldering and assembly of electronics, optical alignment, python-based experimental solutions, physical modelling and research.

## Student Specification

- Understanding of phase transitions, specifically in ice
- Understanding of crystalline structures
- Python programming skills
- Modelling and outside-of-the-box thinking
- Good problem-solving skills and attention to detail

# Radtest Ltd - Electronics Engineering | Oxfordshire | On-site / Hybrid

## Your Contact

Name: Jack Praill
Telephone: 01235 436622
Email: jack.praill@radtest.co.uk

## Organisation Address

Radtest Ltd, D Wing, HQ Building, Thomson Avenue, Harwell, Oxfordshire, OX11 0GD, UK.

## Start and End Date

Flexible

## Working Arrangements

In Office/Hybrid

## Company Description

Radtest Ltd provides a comprehensive offering of radiation testing services and advice to customers in the space, nuclear, high-energy physics and industrial sectors. We have a focus on assessing the suitability of highly integrated devices, such as processors and FPGAs, for use in the natural space radiation environment, but also cover applications from materials used in nuclear reactors to medical scanners and qualifying wafer lots of discrete components.
Whilst primarily a service business, we also manufacture and sell Single-Event Effects (SEE) test systems that can be used to screen components prior to full heavy ion testing in a cyclotron or simply as a risk mitigation technique for low-cost missions. Under the brand SEREEL2, this is a high quality, industrially rugged design of instrument, providing the most stable and reliable laser SEE test system on the market.
Radtest Ltd is an independent SME located in the UK space hub at Harwell.

## Project Title

### Precision Dosimetry Mapping and Operational Optimization of a Professional Cobalt-60 Irradiation Facility and Electronics Laboratory

## Project Brief / Job Description

This project bridges theoretical radiation physics and industrial laboratory management at the Harwell cluster. The student will lead the initial development of a high-fidelity dose-map for an on-site Cobalt-60 cell. This will involve taking dose-rate measurements at different locations within the cell and building a graphical model or simulation tool, establishing a baseline characterisation to support space and nuclear industry testing.
The successful candidate will gain hands-on experience in a professional radiation environment, performing measurements using an ion chamber and UNIDOS electrometer. These measurements will be used to validate and refine computational simulations, ensuring a robust understanding of the radiation field with the cell.
Beyond this activity, the student will act as an "operational consultant," evaluating the laboratory, workshop, and cell environments for efficiency, workflow and safety improvements.
There is significant scope for the student to steer the project, particularly in the consultancy role. In addition to refining coding skills, the student will learn professional protocols including ESD safety, cleanliness, and the fundamentals of radiation effects on semiconductors. This is a unique opportunity to experience the full operational cycle of a specialized SME serving the global space and nuclear sectors.

## Student Specification

- Experience in coding and/or CAD
- Interest in electronics, nuclear and space industry
- Interest in professional laboratory environments, or gaining experience in such an environment
- Reliability and ability to work in small teams
- Interest in how businesses work (operations, organisation, workflow etc)

# Rhizo PTX - Climate Technology Product Manufacturing | Guildford | On-site

## Your Contact

Name: Ash Stott
Email: ash.stott@rhizoptx.com

## Organisation Address

Unit 21a, Surrey Technology Centre, Occam Road, Guildford GU2 7YG

## Start and End Date

Flexible

## Working Arrangements

On site as default.
Working from home is available on request.
Typically, a weekly team meeting in person or remote.
Dress code is smart casual.

## Company Description

RhizoPTX is a deep-tech startup redefining chemical manufacturing through advanced electrochemistry. Our core innovation, the Nitrolyzer, allows for the single-step synthesis of green ammonia at low pressures, moving away from massive, polluting centralized plants to distributed, modular systems. Supported by a world-class ecosystem, including Zinc Ventures, The Royce Institute, NPL, and Innovate UK, we are transitioning from lab-proven breakthroughs to industrial-scale prototypes. We offer a high-impact environment for engineers and scientists eager to solve the "Hard-to-Abate" challenge.

## Project Title

### Data-Driven Performance Analysis and Fluid Dynamics for Green Ammonia Electrolysers

## Project Brief / Job Description

During this 4-week placement, the candidate will bridge the gap between experimental results and theoretical modeling. You will develop Python-based tools to process high-frequency data from our Nitrolyzer test stand and utilize SimScale software to perform fundamental fluid dynamics modeling of our electrochemical cells.

By combining experimental data with simulation results this role can begin to reveal the inner working of the electrolyser "black box". Hence, you will be central to the R&D feedback loop, ensuring our modular hardware is backed by rigorous computational analysis.
Reporting to the Director, and as part of this team, you will be central to the design and development of the next generation of our technology.

Key Responsibilities

- Automated Data Pipelines: Develop Python scripts (Pandas, NumPy, Matplotlib/Seaborn) to clean, process, and visualize time-series data from the Nitrolyzer test stand, focusing on mass balance and energy efficiency.
- CFD Modeling (SimScale): Support fundamental fluid dynamics simulations to analyze gas diffusion within the cathode architecture.
- Correlation Analysis: Compare simulation outputs (e.g., pressure drops, flow velocities and distribution) against experimental data from the test stand to calibrate and refine our digital models.
- Optimization Insights: Identify opportunities for performance enhancement, providing data-backed recommendations for stack geometry.

## Student Specification

Technical Skills (essential):

- Python: Intermediate proficiency in data analysis libraries (Pandas is essential).

Technical Skills (recommended):

- Simulation: Experience with CFD (Computational Fluid Dynamics) tools; specific experience with SimScale or OpenFOAM is desirable but not essential.
- Mathematics: General understanding of fluid mechanics (Reynolds numbers, mass transfer) and basic electrochemistry.

Mindset: A problem-solver motivated by deep-tech decarbonization and the challenges of scaling lab-proven technology into industrial products.

Interests: Passion for renewable energy integration and the transition to a circular nitrogen economy.

## Learning Outcomes

- Strengthened Python expertise in a scientific computing context, moving beyond basic scripts to actionable R&D tools.
- Master the workflow of validating CFD simulations using real-world experimental sensor data.
- Contribute to the thermal and fluidic optimization of a world-first electrochemical ammonia system.

# RNLI - Charity/Rescue Services | Poole | Remote / Hybrid

## Your Contact

Name: Russell Hocken and Abbie Hill
Email: Russell_hocken@rnli.org.uk & Abbie_Hill@rnli.org.uk

## Organisation Address

Royal National Lifeboat Institution
West Quay Road
Poole
BH15 1HZ

## Start and End Date

29/06/2026 - 25/09/2026 (Flexible) Note: This is a 4 week placement.

## Working Arrangements

Fully remote or hybrid

## Company Description

The RNLI is the charity that saves lives at sea. Largely delivered by volunteers, we provide lifeboats and lifeguards around the coast of the UK and Republic of Ireland. We also provide water safety services to keep the public safe whilst using the coastline for work and leisure. The placement is within the data and analytics team, and our purpose is to use data science, analytics, BI and research to deliver insight directly to RNLI decision makers.

## Project Title

### Lifeboat Station Environmental Insight App

## Project Brief /Job Description

The Insight & Analytics and Data Science teams are developing an application that provides analysts with clear, visual environmental insights for lifeboat stations. We currently have a series of Python notebooks that retrieve weather and marine data (e.g., wind, waves, swell, tides) and generate basic visualisations such as wind roses and wave roses.

For this project the student will be transforming the notebooks into a user-friendly application. The app will allow analysts to select a location and automatically receive curated environmental insight through interactive charts, maps, and summary metrics.

With support from Data teams, the placement student will work end-to-end: data processing → visualisation → application UI.

## Student Specification

A student who has a knowledge of Python (including libraries like pandas, matplotlib and streamlit) and data visualisation.

The RNLI uses python and Azure Databricks for our data science environment and app deployment, so skills with those tools would be a benefit.

# Royal Marsden Hospital - Public Health (NHS) | London| Hybrid

## Your Contact

Name: Rollo Moore
Telephone: 020 7808 2508
Email: rollo.moore@nhs.net

## Organisation Address

Radiotherapy Physics dept, Royal Marsden Hospital, SW36JJ

## Start and End Date

Flexible

## Working Arrangements

Hybrid

## Company Description

The Royal Marsden NHS Trust is one of the longest established cancer treatment hospitals and we strive to use techniques and technology to deliver safe and optimal radiotherapeutic treatments. Computational aspects of planning and radiation measurement based checking are two sides of the mainstay of our work. Devices such as arrays of air-filled ionization chambers are an effective means of ascertaining radiotherapy linear accelerator performance. This project is to assess data collected and form a viewer tool on a virtual compute environment CoCalc to extend work in place with previous SEPnet colleagues. Local and remote experiment and computing support is planned.

## Project Title

### Using MPCDataExplorer to Organise PTW Array Beam Data

## Project Brief / Job Description

Curation of a collection of datafiles in preparation for processing by python codebase (“data wrangling”) to organise intercomparison of the relevant derived parameters (“analysis”). Enhancing the existing code (“python coding”) to correlate parameters with existing data. (“statistics”). Visual checks (“presentation”) and interfacing (“exporting”) the reformatted verified data into a separate record database in summary and uniquely traceable way (“communication and audit”).

## Student Specification

- Interest in data handling and radiation science applications.
- Computer literate
- Interest in Data Analysis
- Good attention to detail
- Desire to use scientific skills to benefit patient care
- Awareness of hospital environment considerations.
- Happy to work in a clinical environment (May on occasions see bodily fluid/distressed patients)

# Royal Surrey County Hospital - Public Health (NHS) | Surrey | Hybrid

## Your Contact

Name: Oliver Berry
Telephone: 0148340660
Email: o.berry1@nhs.net

## Organisation Address

Nuclear Medicine, Royal Surrey County Hospital, Egerton Road, Guildford, GU2 7XX
Tel: 01483406708
(Hospital accommodation is available, application required)

## Start and End Date

Flexible

## Working Arrangements

Hybrid

## Company Description

Nuclear Medicine uses radioactive materials to diagnose and treat disease. It is multi-disciplinary, synergising expertise from clinicians, technologists and physicists. This role is based within the Nuclear Medicine Physics section of the Royal Surrey's Medical Physics department.
The Nuclear Medicine team consists of 11 physicists, providing scientific support to 6 clinical nuclear medicine departments and 3 fixed-site PET/CT services in the South East of England. Facilities at the Royal Surrey include 3 SPECT/CT systems (including one with state-of-the-art solid state detector technology) and radionuclide laboratory facilities. We support a further 8 gamma camera systems, including 2 SPECT/CT systems, at other local hospitals. The department also provides scientific support for a number of mobile PET/CT scanners as part of private contracts. We are active in research and development, with current collaborative projects with the University of Surrey and the National Physical Laboratory. There is also an active programme of clinical trial research within the hospital.

## Project Title

### Virtual Clinical Trials in Molecular Radiotherapy

## Project Brief / Job Description

What if we could test new cancer treatments before running real clinical trials? Virtual clinical trials (VCTs) use physics-based computer simulations to model realistic patient populations, treatment delivery, and biological response. This allows researchers to explore new treatment strategies, quantify uncertainty, and identify promising approaches before costly and time-consuming clinical trials are undertaken.

This is particularly relevant in molecular radiotherapy (MRT), where radioactive drugs are used to deliver targeted radiation to cancer cells. Although MRT is widely used in conditions such as thyroid cancer and neuroendocrine tumours, predicting treatment outcomes remains challenging because of large patient-to-patient variability in drug uptake, radiation dose delivery, and biological response. Virtual clinical trials, which combine radiation physics, pharmacokinetics, dosimetry, and statistical modelling, offer a powerful way to investigate how these factors interact and how treatments might be optimised.

In this project, you will work with an existing Python modelling framework to simulate MRT treatments, explore uncertainty propagation, and investigate applications to different radionuclide therapies. The project involves computational modelling, data analysis, and radiation physics, with scope for independent exploration and potentially publishable results

## Student Specification

The successful applicant will have:

- An interest in clinical medical physics
- Reasonable coding experience (ideally Python) and an interest in computational modelling
- A good understanding of nuclear and applied radiation physics
- The ability to work well with other scientists, clinical colleagues and patients;
- A strong desire to apply their knowledge and expertise for the benefit of patients;
- Good time management, self-motivation and attention to detail.
- Prior clinical knowledge is not required — curiosity and good problem-solving skills are key.

# Royal Surrey County Hospital - Public Health (NHS) | Surrey | On-site

## Your Contact

Name: Melanie Cunningham
Telephone: 0148340660
Email: mcunningham1@nhs.net

## Organisation Address

Radiotherapy Physics, Royal Surrey County Hospital, Egerton Road, Guildford GU2 7XX

## Start and End Date

Flexible

## Working Arrangements

In Office. Hospital accommodation is available, application required.

## Company Description

The radiotherapy physics department consists of over 40 healthcare scientists, radiographers, technicians and admin staff who all work to ensure radiotherapy treatments are delivered safely and effectively. This principle is behind all our work which includes

- Commissioning or regular testing treatment machines to ensure they meet performance standards
- Planning patient treatments
- Delivery of brachytherapy services
- Evaluating and introducing new treatment technologies and processes

The department has 7 linear accelerators, 3 CT scanners a number of superficial treatment units and an active brachytherapy service

## Project Title

### Evaluation of automated planning methods for Head and Neck cancer

## Project Brief / Job Description

In addition to spending time in all areas of radiotherapy physics section to gain an understanding of the work performed by a radiotherapy physicist a project will be performed to audit quality of treatment plans.

The project is using Eclipse RapidPlan and Varian Ethos as two novel clinical automated planning systems and ProKnow (a newly introduced nationally system) to compare the radiotherapy plan quality against national standards and each other. This will involve developing workflow and scripts to automatically transfer and populate large datasets of images and dosimetric parameters. Ensuring the data imported is correct and complete This will then allow the treatment planning to be accelerated leading to fast throughput for patients and reducing time to treatment.

## Student Specification

- Computer literate
- Interest in Data Analysis
- Good attention to detail
- Desire to use scientific skills to benefit patient care
- Happy to work in a clinical environment (May on occasions see bodily fluid/distressed patients)

# South West London Integrated Care Board - Health (NHS) | Wimbledon | Hybrid

## Your Contact

Name: Dr Catherine Heffernan
Email: catherine.hefffernan@swlondon.nhs.uk

## Organisation Address

South West London Integrated Care Board
120 The Broadway
Wimbledon
SW19 1RH

## Start and End Date

Flexible (1st June to 30th Sept 2026)

## Working Arrangements

Hybrid with an expectation to do 1-2 days in the office a week

Most of our staff work in the office two days a week and the remaining days working from home or remotely. Most meetings are conducted virtually on MS Teams so the students will be able to engage in team discussions and meetings and will be exposed to what modern day working is like.

## Company Description

NHS South West London Integrated Care Board (ICB) is a statutory organisation bringing together the NHS across the 6 boroughs of Southwest London (Kingston, Richmond, Wandsworth, Sutton, Merton and Croydon). The aim is to improve population health and establish shared priorities for local people. The ICB alongside the integrated care partnership (ICP) make up the SWL integrated care system. We work across the South West London Integrated Care System with local authorities, voluntary sector and healthcare providers, working together to redesign care, improve population health and address the wider determinants of health.

See NHS South West London Integrated Care Board (icb.nhs.uk) for more details NHS South West London Integrated Care Board (icb.nhs.uk)

## Project Title

### Public Health Intelligence/Population Health Management

## Project Brief / Job Description

Population health improvement is key to the work of NHS SWL Integrated Care Board (NHS SWL ICB). This involves reducing the occurrence of ill health, improving health outcomes of the population, improving physical and mental well-being of people in a defined population, delivering appropriate health and social care services and working with communities and partners on the wider determinants of health. To do that, we need good data and intelligence and population health management is a key methodology.

Through this placement, the student would be involved in risk stratification of populations and would undertake predictive system modelling (i.e. modelling of expected outcomes under different scenarios using person level data). This is used to consider the impact on health, finances and sustainability. There will an array of different projects the student can work on and the student will have the opportunity to see how the business intelligence team works and how information is used to improve healthcare services, including reducing waiting lists and making efficiencies.

The student will get a good introduction into how public health works and how the 'back office' of the NHS works. They will also be supported to present their findings to key decision makers and see how their work is taken forward. This will be a great introduction to how public services work should they be interested in working in public services in the future. They will also be supported in seeing how transferable their mathematical and modelling skills are in improving people lives.

## Student Specification

We are looking for an analyst who has an understanding of statistics and may be able to apply their mathematical skills to modelling of preventative measures (sustainable demand management) that can lead to cost-savings to the NHS and improvements in quality of healthcare. The student will be supported in seeing how transferable their mathematical and modelling skills are in improving people lives. This is a 6-8 week placement.

Key skills to include:

- An interest in data
- An interest in public health
- Knowledge of coding/ effective manipulation skills of large amounts of data

# Thames Water - Utilities/Water | Reading | Hybrid

## Your Contact

Name: Leigh Maxfield
Telephone: 07747645761
Email: leigh.maxfield@thameswater.co.uk

## Organisation Address

Clear Water Court, Vastern Road, Reading, RG1 8DB

## Start and End Date

June- August Flexible

## Working Arrangements

Hybrid and some site visits

## Company Description

Thames Water is the UK's largest water and wastewater company. We make a daily difference to our 15 million customers by supplying 2.6 billion litres of water through 32,000 km of pipes, keeping taps flowing and toilets flushing.
We are dedicated to delivering a continuous supply of drinking water to our customers through our infrastructure. The Water Networks Repair and Maintenance teams are responsible for completing repair and maintenance activities on Thames Water's underground water network – pipes and ancillary fittings.At Thames Water, every one of our actions, big and small, matters every day. Water is essential to daily life, and that means our business is always open.

## Project Title

### Thames Water Summer Placement

## Project Brief / Job Description

In this placement the successful candidate would be facilitating an efficiency project and will own from start to finish a short project either in chemicals, H&C, Commercial and/or energy which will demonstrate savings to the business. They will also be exposed to collaborating across Thames Water and the water sector as a whole.
The student will apply specific technical skills and knowledge to deliver solutions in a timely manner within cost and quality standards. The successful student will supply and analyse information to satisfy complex needs of the team and provide explanation or commentary. They will design and deliver an element of a technical solution and assist the team to implement it successfully. As well as Identify developments in the relevant technical

specialism that may improve the business, and suggest initiatives to the team and/or line manager.

However, we take a student centred approach, if you have the behaviours and attitude we are looking for a bespoke project will be made for you and your interests. Previous placements include:

- In process engineering, an intern was able to optimise the build sequence of a capital project at one of our large water treatment sites. This will deliver cost savings, streamline the onsite build, and reduce the likelihood of serious water quality failures.
- A mathematics student in the system optimization team created a model to optimise the prioritisation of transient pressure to aid early detection of leaks
- A biomedical student within our Research Scientists assisted the team with greenhouse gas project and using her knowledge investigated the Polio Virus at Wastewater Treatment sites.
- In capital delivery, an intern produced a reliable financial dashboard that automatically pulls complex project data reducing the need for human interaction. This will ensure month-end processes are quicker and delivered with higher degree of accuracy

## Student Specification

We are super flexible when it comes to students. If you are interested in the water industry, have enthusiasm to work as part of a team, and skills to bring to the table we want to hear from you.

# UKAEA - Fusion/Research/Energy | Oxfordshire | Hybrid

## Organisation Address

UK Atomic Energy Authority, Culham Science Centre, Abingdon, Oxfordshire, OX14 3DB.

The UK Atomic Energy Authority is accessible by train, car and bike from Oxford, Abingdon, and Didcot. Site is 10min from Culham rail station with trains to Didcot and Oxford, also bus service to Abingdon and Cowley.

### Project Title: Development of Duct / Baffle Systems to Reduce the Flux of Neutral Particles Onto the Optical Components

Applications must be made through the following link
https://df5276.github.io/sepnet-2026/am01.html

## Your Contact

Name: Adam Parrott
Email: adam.parrott@ukaea.uk
Telephone: 01235 467875

## Start and End Date

Start dates are flexible. The placement is for 12 weeks

## Working Arrangements

Hybrid – the student can work almost entirely remotely agreeing which days to visit site with their supervisor.

## Company Description

By 2050, the planet could be using twice as much electricity compared to today. Are you interested in joining an organisation where your work will contribute and help to shape the future of the world's energy? If so, our summer placement scheme provides a perfect gateway into a future career in this industry.

Fusion, the process that powers the Sun and Stars, is one of the most promising options for generating the cleaner, carbon-free energy that our world badly needs. UKAEA are at the forefront of realising energy from fusion, working with industry and research partners to deliver the ground-breaking developments that will underpin tomorrow's fusion power stations with the aim of bringing fusion electricity to the grid.

The placement will be within the division of the UKAEA that is most concern with tokamak plasma physics. The student will be supervised by experts in the design of tokamak diagnostic systems (systems that sense the plasma within the tokamak). The members of this group are primarily physicists with experience designing and operating tokamak diagnostic systems.

## Project Brief / Job Description

Many tokamak diagnostic systems utilise optical components. A significant challenge in the design and operation of these optical systems relates to their degradation in the fusion environment, not only due to high energy neutrons, but due to contamination of optical surfaces. Impurity atoms from the tokamak wall, energetic particles from the plasma, and neutral particles from around the plasma edge and divertor will coat the optical components reducing the diagnostic system performance. Within this project, you will help to develop systems to reduce the flux of neutral particles from the plasma onto these optical components. This project will likely require modelling the physics of neutral particle transport, surface physics of particle adsorption and possibly strategies for measuring optical component degradation state. You'll need to decide which modelling approaches to deploy, these may include commercially available software packages or other opensource tools. You will work with other teams who simulate particle transport in the divertor and plasma edge, and those engaged in the design of vacuum systems.

## Student Specification

- Statistical mechanics of high Knudsen number flows and how they may be simulated.
- The different approaches to modelling the behaviour of optical systems. How coatings affect mirrors and optical components, how those effects may be simulated.
- An interested in the design of tokamak diagnostic systems.
- An interest in carrying out physics simulations using either opensource or proprietary software.

# UKAEA - Fusion/Research/Energy | Oxfordshire | Hybrid

## Project Title: Accelerating the Simulation of Rarified Gas Flows to Guide Tokamak Design

Applications must be made through the following link
https://df5276.github.io/sepnet-2026/at01.html

## Your Contact

Name: Adam Parrott
Email: adam.parrott@ukaea.uk
Telephone: 01235 467875

## Start and End Date

Start dates are flexible. The placement is for 12 weeks

## Working Arrangements

Hybrid – the student can work almost entirely remotely agreeing which days to visit site with their supervisor.

## Company Description

By 2050, the planet could be using twice as much electricity compared to today. Are you interested in joining an organisation where your work will contribute and help to shape the future of the world's energy? If so, our summer placement scheme provides a perfect gateway into a future career in this industry.

Fusion, the process that powers the Sun and Stars, is one of the most promising options for generating the cleaner, carbon-free energy that our world badly needs. UKAEA are at the forefront of realising energy from fusion, working with industry and research partners to deliver the ground-breaking developments that will underpin tomorrow's fusion power stations with the aim of bringing fusion electricity to the grid.

## Project Brief / Job Description

The tokamak divertor is a region within the tokamak where energetic particles are exhausted from the plasma. Without careful management of the divertor state, the heat loads could be many times the flux experienced by space craft heat shields during atmospheric re-entry. Such heat loads would rapidly melt the tungsten divertor structure. To reduce these heat loads, high Z effective gases are injected into the divertor region. The presence of these converts the focussed energetic particle flux into EM radiation via stimulated emission and other related processes. However, should the impurities migrate into the plasma core, the resulting emission would lead to rapid cooling of the tokamak plasma resulting in a

damaging disruption event. Not only do the concentration of impurities gases need to be carefully controlled, but also helium ash from nuclear fusion reactions accumulates and valuable tritium fuel that may dwell in the divertor needs to be returned to the plasma core. Reducing the concentration of impurity gases, helium ash and tritium re-cycling is achieved partly by the use of turbo-molecular pumps and cryogenic adsorption panels. This project concerns the simulation methods used by those designing these systems. Challenges the group face include long simulation times, coupling of different methods of physics simulation (e.g. switching from continuum flow modelling to using rarefied gas dynamics simulation methods including Monte Carlo and lattice Boltzmann methods). Through this project, you will understand the challenges the team face and help them with developing their workflows. The topics you are likely to cover include the use of machine learning/AI to accelerate simulations, use of your physics knowledge to help identify how and where to switch the physics simulation methods used, and helping to get workloads running on high performance computing facilities. You'll work between different teams and you may help to couple together the different physics simulations undertaken by different groups. Relatedly, you may work with other SEPnet students who may need to simulate low density flows for diagnostic system design applications.

## Student Specification

Statistical mechanics of high Knudsen number flows and how they may be simulated.

Helpful to have some interest/knowledge in machine learning topics for creating surrogates to accelerate physics simulations.

# UKAEA - Plasma Fusion/Research | Oxfordshire | Hybrid

## Project Title: Cross-code Comparison of Interpretive Core Plasma Modelling for High-performing MAST-U Discharges

Applications can only be received via online application.
The application link is: https://df5276.github.io/sepnet-2026/zs01.html

## Your Contact

Name: Dr. Žiga Štancar
Email: Ziga.Stancar@ukaea.uk

## Organisation Address

UK Atomic Energy Authority, Culham Science Centre, Abingdon, Oxfordshire, OX14 3DB.

The UK Atomic Energy Authority is accessible by train, car and bike from Oxford, Abingdon, and Didcot.

## Start and End Date

01/06/2026 - 30/09/2026 (approximate and/or flexible 12 week placement will be discussed with candidates)

## Working Arrangements

Hybrid 2 or 3 days a week in office

## Company Description

By 2050, the planet could be using twice as much electricity compared to today. Are you interested in joining an organisation where your work will contribute and help to shape the future of the world's energy? If so, our summer placement scheme provides a perfect gateway into a future career in this industry.

Fusion, the process that powers the Sun and Stars, is one of the most promising options for generating the cleaner, carbon-free energy that our world badly needs. UKAEA are at the forefront of realising energy from fusion, working with industry and research partners to deliver the ground-breaking developments that will underpin tomorrow's fusion power stations with the aim of bringing fusion electricity to the grid.

## Project Brief / Job Description

Work with physicists in the dynamic and diverse Plasma Core Team to further the understanding of high-performance plasmas performed at UKAEA's MAST-U spherical tokamak, and the integration of plasma scenarios relevant for future fusion reactors. Within the project you will learn how to use and apply JETTO, a state-of-the-art core plasma transport code, which has been extensively used for analyses of existing fusion experiments and for predicting the evolution of plasmas in future fusion devices such as ITER and UKAEA's pilot plant STEP. The aim of the project is to benchmark JETTO's interpretive modelling capability against TRANSP, a golden standard transport code developed at the Princeton Plasma Physics Laboratory, on a variety of MAST-U discharges from the recent experimental campaign. This will help our effort to run JETTO routinely for MAST-U discharges, aiding our interpretation of the plasma state, facilitating efficient experiment development and optimization, and validate core transport models. As part of your placement, you will receive training and guidance on relevant plasma physics, modelling codes and career development, as well as hands-on support with the assignment from experienced modellers. You will also analyse, visualize, and present the modelling results to the Plasma Core Team, mastering best practices of data analysis and dissemination of scientific results.

## Student Specification

The candidate should demonstrate:

- An interest in plasma physics and wish to contribute to the development of fusion energy.
- Motivation to learn how to use plasma physics modelling tools and work with modellers in the Plasma Core Team.
- Some experience in programming relevant to data analysis and visualization.
- Desire to improve programming, data analysis and presentation skills.
- Understanding of equality, diversity and inclusion principles and commitment to respect these in the workplace.

# un:hurd Music - Music Tech | London | Hybrid

## Your Contact

Name: Giorgio Cerro
Telephone: 07901888303
Email: giorgio@unhurd.co.uk

## Organisation Address

un:hurd music, Tileyard Rd, London N7 9AH

## Start and End Date

4 weeks flexible (start between June/July)

## Working Arrangements

3 days in office but can be flexible Tue-Wed-Thu.

## Company Description

The company - un:hurd music

For years, artists have been 'shooting in the dark' when it comes to their marketing. There's a clear lack of robust, transparent and accessible marketing tools for musicians and artists to use to promote their music.

At un:hurd music, we've created a music-tech platform that uses proprietary technology to turn music data into stronger promotional campaigns for artists. We've created both mobile and web applications which act as the 'digital marketing companion' for independent artists, facilitating data-led promotional campaigns across streaming and social media platforms.

The team

We operate as an agile, high-growth start-up where every contribution matters. You will work alongside an ML Engineer and a Data Lead in a team that manages the entire data lifecycle. This offers you a unique opportunity to learn and hone your skills across the full data stack—from data engineering to advanced Data Science and ML or AI prototyping and deployment

## Project Title

### Playlist Recommendation Engines

## Project Brief / Job Description

We are looking for someone to design and build a prototype that will directly enhance our product data offering. You will focus on unlocking the value of our large, rich and under-utilized datasets (streaming platform numbers, social media data, radio, concert, etc) to solve complex challenges that independent artist face when promoting themselves. You will research and use data techniques adapted to solve the problem at hand: ranging from data visualisation to statistical analysis and state-of-the-art ML models.

- Your exact project will be refined based on your interests and alignment with our roadmap at the time of your arrival. It will target high-impact areas such as:
- Playlist Recommendation Engines: Enhancing our genre-based tools to efficiently weigh features such as mood and language, to surface the best playlists to our users.
- Virality Detection: Analyzing time-series data to detect early growth trends that indicate a track is entering a "virality loop" on streaming or social media platforms.
- Ad-Tech Optimization: Automating our advertising tool to ensure artists present the right creatives to the right audience at the exact right moment.
- Influencer Matching: prototyping our next artist-to-influencer matching tool.

## Student Specification

We are looking for a student who wants to gain applied Data Science experience within a fast-moving start-up environment.
You should be an independent thinker and a proactive problem-solver who is excited by the challenge of building solutions that help independent artists thrive.

You should be capable of taking a problem, identifying the best method for the job (from analytics and data visualisation to ML and AI), and turning it into a concrete solution.

Requirements:
- Strong Python coding skills
- Experience handling and extracting value from complex datasets
- Machine learning experience

Preferred/Bonus:
- Familiarity with version control (Git)
- Experience querying databases (SQL)
- Exposure to cloud computing environments.

# University College London - Higher Education/Research | London | On-site

## Your Contact

Name: Abbie C. Bray
Email: a.bray@ucl.ac.uk

**ONLINE APPLICATION SEE DETAILS BELOW - DEADLINE 10th APRIL 2026**

## Organisation Address

London Centre for Nanotechnology (LCN), 17-19 Gordon Street, London, WC1H 0AH

## Start and End Date

June-September - flexible

## Working Arrangements

Based at University College London UCL on the Bloomsbury campus within the LCN or Physics Department.

## Company Description

The UCL EPSRC Centre for Doctoral Training in Quantum Communication and Quantum Computation aims to train the next generation of leaders in Quantum Technologies research. Our work brings together a team over 40 academic experts from 5 university departments, including Physics and Astronomy, Computer Science, Electrical and Electronic Engineering, the London Centre for Nanotechnology, and Chemistry. Quantum technologies involve the control and manipulation of quantum states to achieve results not possible with classical matter; they promise a transformation of measurement, communication and computation. As breakthroughs in quantum technologies move out of the lab and into industrial applications, the highly skilled researchers who will be the future leaders in this field must be equipped to function in a complex research and engineering landscape where quantum physics meets cryptography, complexity and information theory, devices, materials, software, and hardware engineering.

## Project Title: Quantum Technologies Research Summer Placement Scheme

## Project Brief / Job Description

Are you looking for a research placement or internship this summer and considering further study in science & technology after your undergraduate degree?

UCL's Quantum Technologies undergraduate summer placement program is now open for applications! Come to UCL this summer for a paid 8-week research project, working with academics and postgraduate researchers from UCL's Centre for Doctoral Training in

Quantum Communications and Quantum Computation.
These placements are an excellent way to gain experience working in a lab on a research project and obtain research experience during the summer months. Each placement takes place between June and August 2025 and includes a £3437 bursary for living expenses and other costs.

## About Us

The UCL EPSRC Centre for Doctoral Training in Quantum Communications and Quantum Computation aims to train the next generation of leaders in Quantum Technologies research. Our work brings together a team of over 40 academic experts from 5 university departments, including Physics and Astronomy, Computer Science, Electrical and Electronic Engineering, the London Centre for Nanotechnology, and Chemistry.
Quantum technologies involve the control and manipulation of quantum states to achieve results not possible with classical matter; they promise a transformation of measurement, communication and computation. As breakthroughs in quantum technologies move out of the lab and into industrial applications, the highly skilled researchers who will be the future leaders in this field must be equipped to function in a complex research and engineering landscape where quantum physics meets cryptography, complexity and information theory, devices, materials, software, and hardware engineering.

YOU MUST APPLY VIA THE WEBSITE:
Find out more and apply online through our website here.

## Student Specification

To apply for these placements:

- You must be enrolled on an undergraduate degree programme at a UK university (e.g. BSc, BEng, MSci, MEng)
- You must currently either be in the second year of a three-year course or the third year of a four-year course; i.e your penultimate year of study.
- You cannot also apply for admission to the EPSRC Centre for Doctoral Training's PhD programme this year
- No previous experience is required: we welcome applications from students who have not done an internship or research placement before

We particularly welcome applications from women, people with disabilities, and candidates from minority ethnic and socially disadvantaged groups as they are under-represented within STEM disciplines. UCL is committed to equality of opportunity, supports and encourages under-represented groups, and values diversity.

**PLEASE MAKE SURE WHEN APPLYING YOU PUT (SEPnet) against your name**

# University Hospital College London - NHS/Medical Healthcare | London | Hybrid

## Your Contact

Name: James Leighs
Email: james.leighs1@nhs.net
Co-project Supervisor - Alison Walley

## Organisation Address

Clinical Scientific Computing (CSC),, Medical Physics and Biomedical Engineering (MPBE), EGA Wing,, Level -2 (Basement), UCL Hospitals, 235 Euston Road, London NW1 2BU

## Start and End Date

Flexible (This is likely to be a 4-week placement)

## Working Arrangements

Hybrid

## Company Description

University College London Hospitals NHS Foundation Trust (UCLH) is one of the most complex NHS trusts in the UK, serving a large and diverse population. We provide academically led acute and specialist services, to people from the local area, from throughout the United Kingdom and overseas. Our vision is to deliver top-quality patient care, excellent education, and world-class research.

The Clinical Scientific Computing (CSC) Section is one of the largest in the country, and provides a wide array of services to clinical, scientific, and research teams, predominantly throughout UCLH. You'll be joining us at an exciting time as our team is evolving to meet the growing demand for digital and data-driven healthcare, working closely with clinical and technical stakeholders to develop and deliver a full range of high-quality scientific computing services. We are expanding our research support and delivery as we grow our programming and data science function within the team. In addition, we currently support a wide range of areas, from bespoke clinical systems and infrastructure to regulated medical device software, with a growing emphasis on data science, interoperability, and automation. One such area that we support and are starting to collaborate more closely with is the UCLH Proton Beam Therapy (PBT) unit, a world class cancer treatment centre and one of only two such services in the NHS. This project will bring together staff from both CSC and PBT so the successful candidate has a great opportunity to get experience of both fields.

If you're motivated by solving real world healthcare challenges, enjoy thinking big, diving into technical detail, and value emotional intelligence and communication as much as scientific skills, please get in touch.

## Project Title: Development of a Web-based Database Platform to Support Quality Assurance Processes in the Proton Beam Therapy (PBT) Unit

## Project Brief / Job Description

Proton Beam Therapy (PBT) at UCLH relies on a legacy Microsoft Access database to store quality assurance (QA) results and track specialist equipment. However, Access is no longer supported within the Trust and cannot meet the long-term needs of a cutting-edge clinical service. Commercial alternatives such as QATrack+ and RadMachine offer partial solutions, but none provide the full functionality required and current funding does not allow for their full adoption. This creates a unique opportunity for an ambitious student to design and develop a bespoke system from the ground up.

This project aims to build a modern SQL-based database with an intuitive web interface that enables physicists, technicians, and clinicians to efficiently input, view, and analyse QA data. The system will support universal access across the Trust and allow advanced interrogation of data for visualisation, trending, and long-term performance tracking.

Students will gain hands-on experience with real-world clinical workflows, database design, and interface development, working directly with the Head of QA and a team of Clinical Scientists in Clinical Scientific Computing experienced in coding and database systems. This is a rare chance to create a tool that will directly support patient safety, scientific innovation, and the future of proton therapy.

## Student Specification

Essential

- Interest in software development and data science
- Basic experience with programming (e.g. Python)
- Quick and self-driven learner, with a willingness to learn SQL and web-based interface development
- Good problem-solving skills and attention to detail
- Ability to work independently and collaborate within a multidisciplinary team
- Interest in medical physics, healthcare technology, or data visualisation
- Enthusiasm for building real tools that will be used in a clinical environment

Desirable

- Experience with databases (e.g. postgres)
- Familiarity with web frameworks (e.g., Flask, Django)
- Desire to develop a career in clinical scientific computing and/or medical physics

# University Hospital Southampton - NHS/Medical Healthcare | Southampton | Remote

## Your Contact

Name: Riannach Semple
Telephone: 023 8120 6109
Email: Riannach.Semple@uhs.nhs.uk

## Organisation Address

Scientific Computing, Imaging Physics Group, Medical Physics, University Hospital Southampton NHS FT
Southampton General Hospital, Tremona Road, Southampton SO16 6YD

## Start and End Date

6th July  flexible This placement is likely to be a 4 week placement with a possibility of extension tbc

## Working Arrangements

Fully remote with the option of coming on site occasionally

## Company Description

We are the University Hospital Southampton Scientific Computing Team. Situated within Medical Physics we support scientific computing projects within Radiotherapy, Imaging Physics (such as Nuclear Medicine and MRI) and also working with teams in the broader Trust; working with the oesophageal cancer MDT for service evaluation, or the Southampton Oxford Retrieval Team (SORT) to build the SORT App (available on the app store), and many more.
We provide end-to-end specialist clinical software design, development and support, and with an ISO13485 accredited QMS, as well as being involved in scientific research and research imaging data management. The scientific computing team develops bespoke scientific software for clinical and research uses, usually working with imaging data from multiple modalities as well as some non-imaging datasets. Our recent developments include an automated dementia imaging reporting tool for Nuclear Medicine imaging data, an iOS application for calculating paediatric drug doses, and a statistical reporting tool for neonatal and infant MR (Magnetic Resonance) spectroscopy data. We work closely with colleagues within our imaging physics group (representing Magnetic Resonance Imaging, Nuclear Medicine and Positron Emission Tomography and Neurological Measurements), and with other groups including UHS (University Hospital Southampton) Digital, groups at the University of Southampton, Research & Development hubs such as the Biomedical Research Centre and research active specialities such as neurology and oncology.

## Project Title

### Clinical Scientific Computing to Improve Image Processing Workflows Through Increased Levels of Automation

## Project Brief / Job Description

The successful candidate will work as part of the Scientific Computing team to develop prototype solutions to enable increasing levels of automation in our current clinical image processing applications and workflows. Depending on the candidate's areas of expertise and interests, this could involve a variety of areas of work including medical image analysis, web development and databases or monitoring and measurement of our current applications. The applicant can choose to work on either a radiotherapy or imaging physics focused project.

## Student Specification

- A quick and disciplined self-learner - the placement will require the ability to find and implement solutions to computing and programming issues that the student has not encountered previously
- Programming experience in Python – it is important that the applicant has at least some experience of programming in Python and the willing to complete some preparatory training to ensure they get the most out of the placement
- Experience in image processing/Machine or Deep learning, ideally in Python, would also be desirable
- The ability to work well with other scientists
- A strong desire to apply their knowledge and expertise for the benefit of patients
- Good time management, self-motivation, and attention to detail.